6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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1 Introduction

Clouds are a major component in regulating the Earth's radiative budget. However, they exhibit the largest spatial and temporal variability compared to gasses and aerosols (int). Additionally, the water, ice, and mixed-phase hydrometeors within clouds can vary significantly. Because of this, clouds remain poorly understood, particularly due to data gaps in regions sensitive to climate change variability, such as the Mediterranean basin. (Marinou et al.). In general, there are few studies on clouds using ground-based remote sensing measurements in Europe, with many located in Northern Europe (Stefan Kneifel et al.; Tatiana Nomokonova et al.; Bühl et al.; Peggy Achtert et al.). In the western Mediterranean basin, the most valuable database of cloud remote sensing measurements is provided by the AGORA ACTRIS CCRES station (Granada station), in the southeast of the Iberian Peninsula. Clouds in this region form through various mechanisms, including thermodynamic conditions, Sahara dust intrusions Casquero-Vera et al., and cut-off lows (COLs) systems, which create environments conducive to storms. These complex formation mechanisms pose significant challenges for accurately representing these clouds in atmospheric models. Consequently, there is a critical need for enhanced cloud analysis and data collection in the Mediterranean region.

Hence, in order to improve cloud analysis, the AGORA (Andalusian Global ObseRvatory of the Atmosphere) station has been part of Cloudnet since XXX and was integrated into ACTRIS CCRES (where ACTRIS stands for Aerosols, Clouds, and Trace gasses Research InfraStructure, and CCRES for Centre for Cloud Remote Sensing) in 2023 as a new facility. The Cloudnet processing chain (J. A. et al.) performs extensive quality assurance on raw instruments for providing high-level products, such as target classification, liquid water path, and droplet effective radius. This chain standardizes data processing within the cloud remote sensing community. Their products are derived using a synergistic approach, integrating vertically pointing measurements from ground-based instruments like Doppler cloud radars, microwave radiometers, and ceilometers. Specifically, the target classification product, generated through these three instruments, has been utilized in numerous studies for cloud typing and multi-year analysis. For example, Stefan Kneifel et al. used this product in the German Alps to analyze ice clouds and precipitation variability at a high-altitude location. The authors found diurnal cycles in summer related to convective clouds, while other months are influenced by large-scale advection of clouds and precipitation. Similarly, Tatiana Nomokonova et al. investigated cloud profiles at the Ny-Ålesund station in the Arctic, noting a maximum cloud occurrence and higher cloud base in summer, differing from satellite observations. Multi-layer and single-layer clouds showed similar frequencies, with mixed-phase clouds being the most frequent single-layer cloud. These mixed-phase clouds exhibited higher thickness but lower ice water content compared to ice clouds. In contrast, Răzvan Pîrloagă et al. studied cloud profiling at the Bucharest–Magurele site, northeast of the Granada station, finding ice clouds to be the most frequent, unlike in Arctic studies. These studies rely on cloud classification methods that categorize clouds based on time or profile into different categories, e.g. liquid clouds, ice clouds and mixed-phase clouds. However, this approach can result in inhomogeneous cloud structures by classifying different cloud types within the same cloud, potentially skewing cloud property statistics and complicating further analysis.

This particular feature can produce high correlations between cloud types, significantly affecting cloud property statistics at some stations. This makes analyzing cloud properties challenging, as the method influences the values, which should not depend on the classification approach. Our study aims to improve cloud profiling and its physical properties using Cloudnet post-processed data by applying a novel cloud classification method. This new approach was motivated by Bühl et al., which deals with the inhomogeneity on cloud classification incorporating a minimum time between profiles for changing from one category to another (e.g. ice to mixed-phase), avoiding sharp changes of cloud types within the same cloud. Nevertheless, this method can also break the cloud into different types depending on the time interval. Thus, we introduce a method based on identifying groups of connected hydrometeors, ensuring homogeneous classification and low correlation among cloud types. Therefore, we explore the advantages to utilize this new approach in relation to the time-based classification. Furthermore, this method were used to evaluate the annual, seasonal, and daily cycles of cloud macrophysics (cloud base, top, and thickness) and microphysics (LWC, IWC, r_{liq}, and r_{ice}). These cloud properties were retrieved using Cloudnet algorithms based on A. S. Frisch et al.; Robin J. Hogan et al.; Hannes Griesche et al.; Julien Delanoë et al..

The experimental site, datasets, instruments and products are described in Section 2. The proposed new approach on cloud classification algorithm and the cloud assessment are presented in Section 4. Section 5 analyzes the annual, seasonal, and daily cycle of cloud types, as well as the base and top height, and thickness. In addition, this section discusses the liquid and ice properties variability for different clouds. Finally, Section 6 summarizes the main findings and discusses their importance to improve further cloud statistics analysis.

2 Experimental site and Instrumentation

The AGORA ACTRIS-CCRES station (37°2’E, 3°6’S, 680 masl) is located in the Southern of the Iberian Peninsula being one the most meridional CLOUDNET stations. [en este párrafo tienes que describir la climatología de la estación porque será relevante para la estadística de nubes.]

2.0.1 Instrumentation

RPG-HATPRO G2 microwave radiometer has been employed, featuring 11 channels around the water vapor absorption band (23.8 GHz), 9 channels around the oxygen absorption band (52 GHz), and 10 channels around the liquid water absorption band (31.4 GHz). This instrument measures brightness temperature with a resolution of 0.3-0.4 K, which is subsequently used to calculate the liquid water path (LWP) and derive the liquid water content (LWC) for liquid clouds.

A CHM15k Nimbus ceilometer operating at a power of 8.4 μJ and wavelength of 1064 nm has been used to provide the attenuated backscattering coefficient of aerosols and cloud droplets. This instrument is used for estimating cloud base height and detecting aerosol layers due to the sensitivity of this wavelength to small particles such as aerosols and water droplets.

A dual polarimetric RPG-FMCW 94 GHz Doppler cloud radar were installed at the station in 2018. It measures the Doppler velocity spectrum (DVS) for horizontal and vertical linear polarizations at 94 GHz. Unlike traditional single polarization measurements that retrieve only reflectivity (Z), mean Doppler velocity (ν_D), and spectral width (S_ω), this advanced radar also derives the linear depolarization ratio (LDR), enabling the detection of the melting layer, aerosols, and insects as described by Hogan and O'Connor. Table 1 provides a summary of the AGORA instrumentation, detailing operational frequencies or wavelengths, type of measurements, products, measurement ranges, data collection intervals, and references with detailed descriptions.

Table 1. Summary of instruments and their data products.

Instrument	Bands / wavelengths	Measured	Products	Range	Time	References
94 GHz Cloud Doppler Radar	94 GHz (W-band)	Doppler velocity spectrum (DVS)	ν_D , Z, S_w	[13,51] m	[1.1, 3.6] s	Nils Küchler et al.
MWR RPG HATPRO	22-31 GHz (K-band) 51-58 GHz (V-band)	Brightness Temperature	q and T LWP	[10, 300] m -	[2, 2.5] min 2 s	Francisco Navas-Guzmán et al.
CHM15k Nimbus Ceilometer	1064 nm		β	15 m	15 s	A. Cazorla et al.

This range varies with the height and depends on the chirp table configuration.
It also depends on the chirp type.
Humidity and temperature profiles have a range varying from 10 to 150 m in the first 4.4 km and from 50 to 300 m between 4.4 and 10 km.

Note that the range and time resolution of the cloud radar vary in altitude and time. It depends on the radar chirp table configuration, where the time resolution, the maximum unambiguous velocity (Nyquist velocity), and the maximum range are defined. These three variables are interdependent and can be strategically adjusted.

3 Cloudnet database

An extensive dataset spanning the years 2018 to 2023 was utilized in this study. Specifically, high-level post-processed data from Cloudnet (J. A. et al.). Figure 1 illustrates the monthly Cloudnet high level products available for the AGORA ACTRIS-CCRES station. The availability of these data depends on simultaneous vertically pointing measurements of the microwave radiometer, ceilometer, Doppler cloud radar, as well as the meteorological model products - the ECMWF model (acronym for European Centre for Medium-Range Weather Forecasts). This figure shows data availability in the analyzed period. The years 2018 and 2020 are the years with the least and greatest amount of data. The white bars denote periods with missing data due to instrument maintenance, technical issues, and scanning measurements. Despite drawbacks, CLOUDNET instrumentation has been continuously operated since 2018.

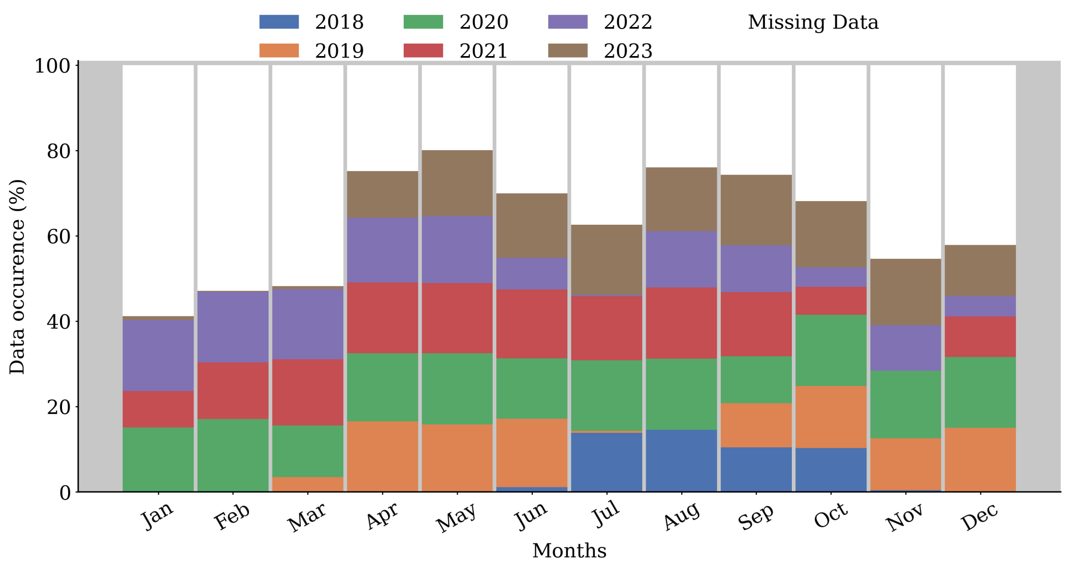


Figure 1. High level post-processed data availability per month at the AGORA ACTRIS CCRES station. This dataset represents more than half a decade of synergic ground-based measurements at this station. Each color represents the data availability for different years.

It is worthy to note that this dataset represents a database for cloud statistics, providing comprehensive data for cloud analysis over an extended period. Notably, it stands as the only cloud remote sensing dataset available in Spain, making it an invaluable resource for regional climate studies. The ACTRIS-CCRES Granada site is uniquely positioned as one of the few locations dedicated to cloud studies in a warm region that is frequently affected by synoptic conditions, Sahara dust intrusion, and thermodynamic conditions. Marinou et al.; F. J. Bello-Millán et al.; Casquero-Vera et al.. This dataset contributes significantly to filling the gap in cloud data for this climate variability hotspot, enhancing our understanding of cloud dynamics and aiding in the improvement of predictive models and climate studies in these critical areas.

3.0.1 Data processing

Instrument datasets are processed by Cloudnet processing chain (J. A. et al.) as follows:

1. Set a common grid with a temporal resolution of 30 s and spatial resolution according with the radar (see table XXX), respectively. The regrinding involved interpolating measurements onto a height grid and was applied to every inter-annual profiling.

2. Corrections: reflectivity gas and liquid attenuation, along with error estimation and quality flag assignment. Specifically, liquid attenuation corrections on radar reflectivity are crucial during raining events, as rain droplets can attenuate a significant fraction of the radar signal (reference XXX).

3. Target classification product. This product classifies atmospheric particles into different categories by assigning an integer ranging from 0 to 10 at each pixel in the Cloudnet processed grid. Categorizes are as follows:: clear sky (0), droplets (1), drizzle or rain (2), drizzle & droplets (3), ice (4), ice & droplets (5), melting ice (6), melting & droplets (7), aerosol (8), insects (9), aerosol & insect (10). These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature (T_w) profiles are used to determine the 0°C isotherm and establish warm and cold regions to identify liquid and ice hydrometeors.

4. Liquid droplet boundaries: lidar attenuated backscatter coefficient (β), and radar reflectivity are used. The first is sensitive to small liquid droplets present in cloud bases, as the second are able to penetrate clouds and detect their tops.

5. Liquid layers respect a threshold of T < -40 °C, since it is not possible to keep water in liquid phase below that temperature. Moreover, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echoes analysis once particle boundaries were already determined. Therefore, falling cold particles were just classified as ice, without distinguishing ice from precipitating ice.

6. Melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 s^{-1} within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on Hogan and O'Connor.

7. The liquid water content (LWC) retrieval assumes a log-normal distribution for warm cloud droplets lifting adiabatically from its base. Its relation depends on atmospheric temperature (T) and pressure (P). This method incurs a relative uncertainty of 15% as shown by A. S. Frisch et al.. At our station, T and P values utilized by the LWC retrieval are provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). The retrieval equation is applied exclusively to liquid pixels within cloud boundaries, which are determined by Doppler radar reflectivity (Z) and ceilometer-lidar attenuated backscatter coefficient. The liquid pixels, as ice, insect and aerosol pixels are given by the cloudnet target

classification product (Hogan and O'Connor). Additionally, LWC values are scaled to align with the Liquid Water Path (LWP) derived from the microwave radiometer. The liquid cloud droplet effective radius (r_{liq}) is retrieved by assuming a log-normal size distribution for warm cloud droplets. This method is insensitive to droplet number concentration and the logarithmic spread of the distribution (S. Frisch et al.). Consequently, its primary source of uncertainty is the reflectivity factor error, resulting in a relative error of 15% for r_{liq} retrievals. This relationship is applied exclusively to liquid pixels, where necessary input data, assumptions, relative errors and the studies from which these products are sourced are indicated in Table 2).

8. The ice water content (IWC) retrieval employs an empirical equation as determined by Robin J. Hogan et al., known as the Z-IWC-T relation, which depends on radar reflectivity and temperature. Compared to other microphysical variables, IWC retrievals has the largest relative uncertainties. For the W-band, the relative error is +55% to -35% for temperatures within the range of -20°C to -10°C, and +90% to -47% for T < -40°C. While temperature is a significant parameter in IWC error estimation, Pablo Saavedra Garfias et al. demonstrated that the primary source of uncertainty is the reflectivity factor error. A typical reflectivity error of 3% can result in a 12% error in the ice water path (IWP), which is the integration of IWC over the atmospheric column. Therefore, errors in IWC can be relatively large. The Z-IWC-T relation is applied only to ice pixels within cloud boundaries. Next, the ice cloud particle effective radius (r_{ice}) is obtained following Julien Delanoë et al., who assumed the same empirical relation for IWC and the visible extinction coefficient (α) derived by Robin J. Hogan et al.. This approach, termed the Z-T relation, depends on both T and Z, and has aThis approach, termed the Z-T relation, depends on both T and Z, and has a relative uncertainty of 50% of r_{ice} (Hannes Griesche et al.). In general, these high relative errors emphasize how challenging it is to retrieve ice properties accurately (see Table 2)

Despite the convenience of using Cloudnet products, target classification at the Granada showed instances of misclassification involving ice, melting, and rain pixels. Figure 2 illustrates a thinner and lower precipitable ice cloud present between 08:00 and 20:00. However, there is no evidence of cloud in the attenuated backscattering and radar reflectivity. These misclassifications stem from inaccurate representation of temperature profiles by models in this region, which subsequently affect cloud classification accuracy. To mitigate these issues and prevent misinterpretation of future cloud statistics, we refined the classification algorithm (see section 4) to better handle these cases.

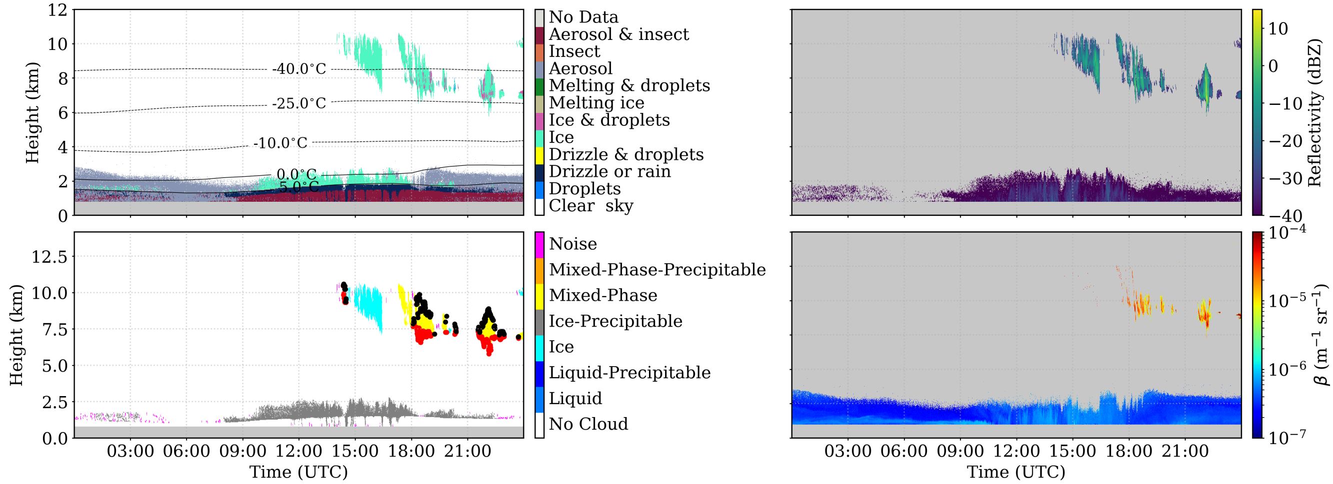


Figure 2. A case study of Cloudnet hydrometeor misclassification. (a) Target classification product; (b) Radar reflectivity factor from Doppler cloud radar; (c) Attenuated backscattering coefficient from ceilometer. No evidence of clouds in the ABL can be identified by the instruments.

Table 2. Cloudnet microphysics products used in this study, their inputs and assumptions

Cloudnet products	Input role	σ_{re}/r_e	References
Liquid water content (LWC)	Z and β: cloud base and top identification T, P, assuming an adiabatic lifting MWR or CDR LWP to scale LWC	15%	Frisch et al. (1998)
Ice water content (IWC)	Z and T profiles	-35% to 90%	Robin J. Hogan et al. Julien Delanoë et al.
Droplet effective radius (r_{liq})	Z, and N and σ of lognormal PSD	15%	S. Frisch et al.
Ice effective radius (r_{ice})	Z and T	50%	Hannes Griesche et al.

4 Methodology

4.1 Cloud assessment by hydrometeor grouping

Target classification product is used to assess cloud type. This product gives the category of pixels in height per time in the Cloudnet grid, as already discussed in section 3.0.1. Thus, the pixels classified as hydrometeors (at a particular height in time) such as Droplets, Ice and Drizzle & Droplets are used to classify clouds and precipitating clouds.

In this study, a cloud is defined as a group of pixels identified by Cloudnet as a cloud with at least 100 pixels (see Figure 3). To do so, a cloud mask is generated from the CLOUDNET target classification. Cloud pixels separated only by two pixels in any direction are considered part of the same cloud. This is performed by a pixel dilatation/contraction technique explained in appendix B. Once the cloud pixels are grouped, each cloud must have more than 100 pixels to be considered (without considering rain such as Drizzle or Rain pixels). Otherwise, it is deemed as an unknown type (likely noise) as can be seen for the magenta pixels in the left-bottom panel of Figure 2.

Then, clouds are classified in six types: liquid, ice, mixed-phase and precipitating liquid, precipitating ice, and precipitating mixed-phase. To do so, the hydrometeor percentages are calculated for each cloud, (shown in figure 3): if a cloud comprised more than 70% of Droplets and Drizzle plus Droplets, it is classified as a liquid cloud (Liquid criteria). If it contained at least 10 connected rain pixels, it is further classified as a precipitating liquid cloud (Rainy criteria); Similarly, if the previous verification was not satisfied, if this group contains more than 90% of ice or less than 10% of Droplets plus Ice & droplets, then it is classified as an ice cloud (Ice criteria). Next, the same rain verification as in the liquid case is done to classify this group as purely ice clouds or precipitating ice clouds. Otherwise, clouds are classified as mixed-phase or precipitating mixed-phase through the same rain verification as other clouds. In relation to precipitating ice and precipitating mixed-phase clouds, an extensive analysis of individual case studies revealed that Drizzle & droplets only appear below the melting layer together with Drizzle & rain droplets. Therefore, to avoid an underestimation of cloud base height due to drizzle, we removed these pixels for these clouds since they are probably rain.

Many recent studies have been employed the algorithm of cloud classification by time (i.e., by profile) with the aim of classify clouds (Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.) and analyze their statistics. However, this algorithm admits different cloud typing inside the same cloud as a whole, assigning similar physical properties for different clouds. In the appendix A, we describe this algorithm, adapting the criteria used by Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.. In another way,

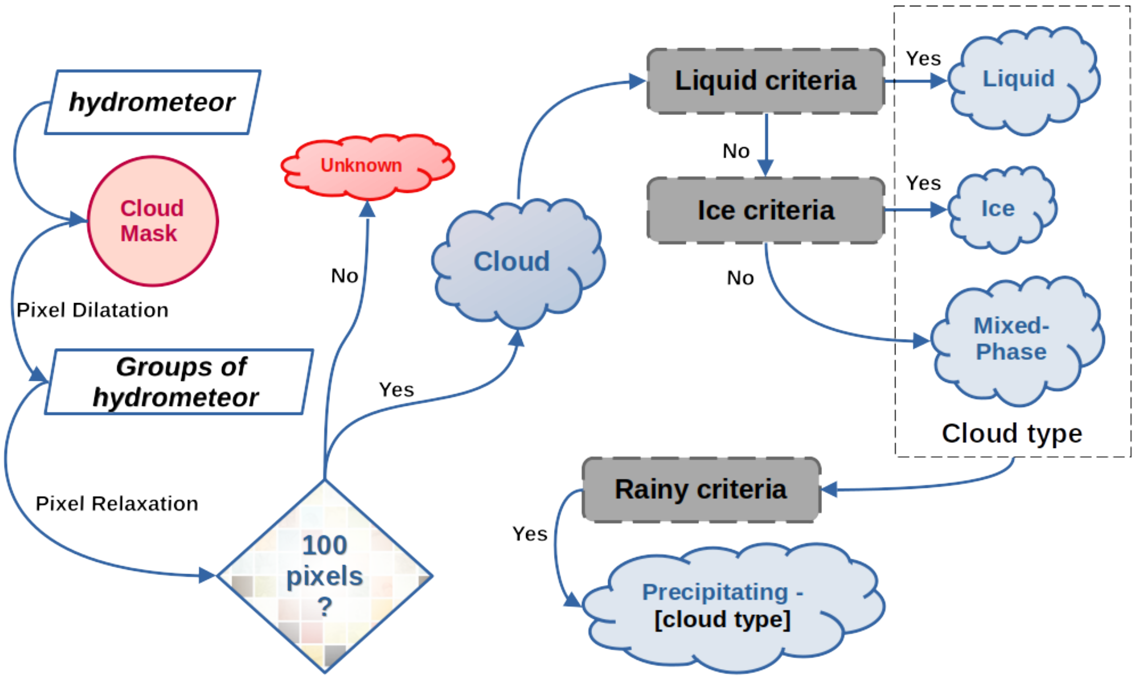


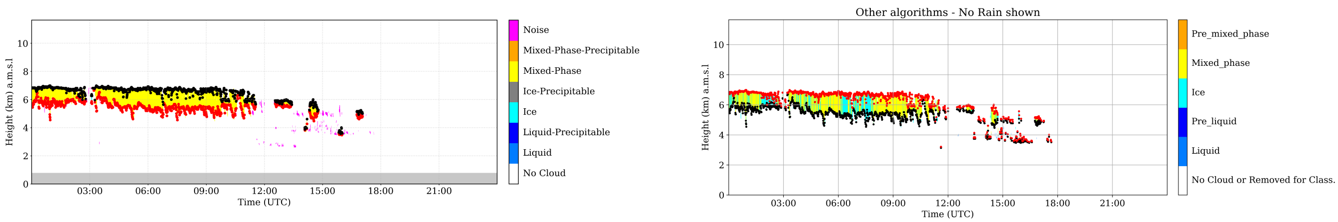
Figure 3. Scheme of the cloud classification algorithm by hydrometeor grouping

After the cloud classification, each cloud is then classified as single or multi-layer over time. Initially, time intervals where clouds overlap are identified. Overleaped pieces of clouds were classified as multi-layer, whereas non-overlapping pieces in principle can be classified as single layer. We then analyzed cloud profiles for each cloud individually, acknowledging that a cloud can exhibit multiple layers with itself. If a cloud had a second layer separated by only one pixel at some time interval, that interval is identified as containing multi-layer. Otherwise, the clouds at that interval are classified as a single layer. For the single layer case, cloud properties were calculated as follows: the cloud base and top height is determined by the first and last height pixels within the cloud layer, respectively. Cloud thickness is obtained by the difference between cloud base and top height pixels.

Taking into account ice, melting and rain hydrometeor misclassification around mid-day as discussed in section 3.0.1, which can result in low level ice and precipitating ice clouds misclassification, these cases were filtered. Basically, all these clouds which were identified with mean cloud base below 4 km (above mean sea level) and average cloud thickness below 700 meters were reclassified as noise. These values were determined empirically by studying many cases separately. Finally, for those cases with no clouds, the time interval is classified as clear sky condition.

4.2 Cloud assessment

A comparative analysis between the two classification algorithms were conducted in order to evaluate their applicability for further analysis. Here, we aim to provide insights into the correlation on cloud statistics between different cloud categories generated by these methods. The findings from this comparison will inform the selection of the most appropriate method for future atmospheric studies and cloud analysis. In this framework, Figure 4 shows a case study on 6 May of 2023 in Granada comparing cloud classification by time and by hydrometeor grouping for a mixed phase cloud, where red and black dots represent the cloud base and top, respectively. As expected, the classification by time on the right panel leads to the fragmentation of clouds into different types. In this case, the same clouds are being classified as ice and mixed-phase by time. On the other hand, in the left panel, the classification by grouping preserves the homogeneity of clouds and classify it as mixed-phase only, resulting in a more realistic classification.



(a) Cloud classification using the new method.

(b) Cloud classification using the old method.

Figure 4. Temporary figure (needs target classification). Comparison of cloud classification methods. The left panel illustrates the new method, which preserves cloud homogeneity and provides a more realistic classification. The right panel shows the old method used in recent studies, where cloud classification is time-based.

As a consequence, the time-based method of cloud classification can generate high correlations between cloud categories in terms of frequency of occurrence and cloud properties. In relation to the cloud occurrence, the right panel of Figure 4 suggests that both ice and mixed phase clouds will have similar daily cloud occurrence and average cloud base. However, this is occurring because of the method of classification, which could affect further cloud statistics. To prove that, the 6-year dataset at the Granada station was used to compute the daily average IWP for ice and mixed-phase clouds using the classification by time. Figure 5 shows the closeness daily ice amount between these two clouds (mixed phase in above panel and ice cloud in the bottom panel) and the similar pattern suggests a correlation generated by the cloud classification. It is worthy to note that this section is not focused on the analysis of IWP, this will be done in section 5.3. Instead of this, here we centered on the correlation generated by the classification algorithms. Therefore, it was clearly observed that high correlation among cloud categories can generate similar physical properties among them, resulting in a misinterpretation of the results. It proceeds because it is difficult to attribute similar properties between clouds (formed by different mechanisms) as a realistic phenomenon that happens in nature or attribute to the method of classification.

Therefore, in order to quantify the correlation between cloud categories generated by these two algorithms of classification, we calculate the daily cloud occurrence to assess the correlation between cloud types. Figure 6 shows the pearson correlation coefficients of daily cloud occurrence between cloud types for the classification by hydrometeor grouping on the left, and for the classification by time on the right. As can be seen, the new proposed method demonstrates generally weaker correlations among cloud types, with most values clustering closer to zero. For example, the highest correlations observed were 0.19

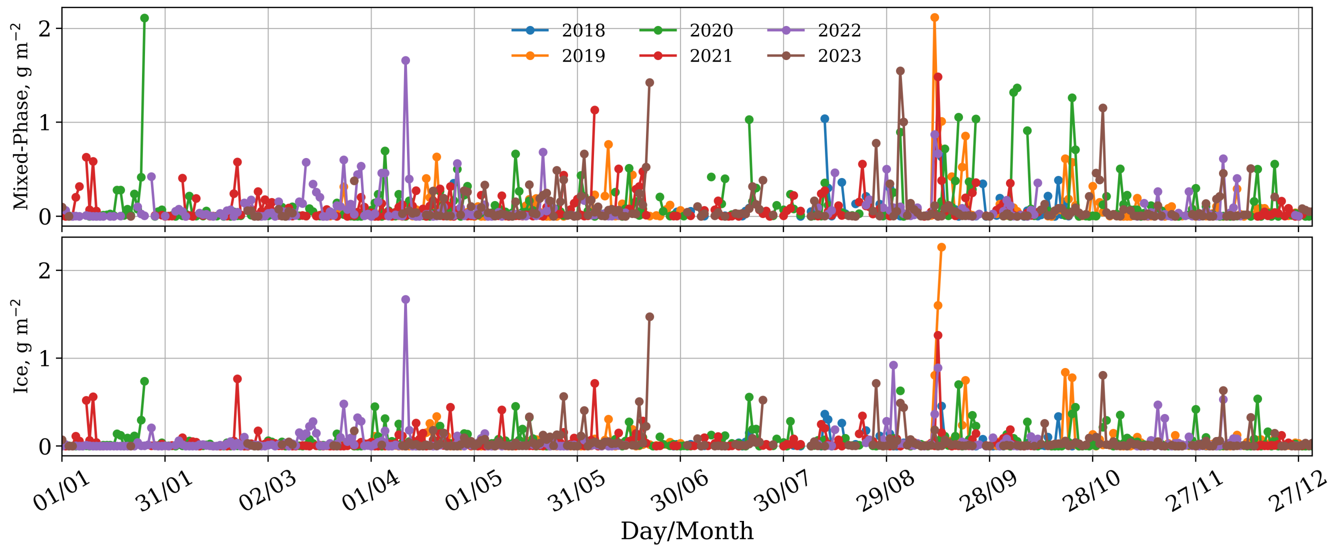


Figure 5. Time series of daily Mixed-Phase (top) and Ice (bottom) cloud ice water path from 2018 to 2023. Each year is represented by a different colored line.

between Liquid and Liquid-Precipitable clouds and 0.17 between Liquid and Mixed-Phase-Precipitable clouds, indicating relatively weak relationships. Additionally, some negligible negative correlations were observed, such as between Ice and Mixed-Phase-Precipitable clouds (-0.1). This suggests that the new method may provide a clearer distinction between cloud types, which is advantageous for analyses requiring precise classification. In addition, computed cloud properties will be not dependent on the method as they must be, avoiding some misinterpretation of the results.

In contrast, the classification by time exhibited stronger and more pronounced correlations. Notably, the correlation between Ice and Mixed-Phase clouds was 0.54, and between Liquid and Mixed-Phase clouds was 0.44, highlighting significant interdependencies between these cloud types. Furthermore, the time-based method showed higher correlations between Liquid and Liquid-Precipitable (0.34), Ice and Mixed-Phase-Precipitable (0.35), and Liquid-Precipitable and Mixed-Phase-Precipitable (0.46), indicating a less distinct separation between cloud types. These higher correlation values are a consequence of the classification scheme defined in this method. The cloud classification by time can fragment them, allowing tagging cloud profiles in different cloud categories, while they are still inside the same cloud structure. Overall, the comparison indicates that the new method, with its generally weaker correlations, may be better suited for distinguishing different cloud types. These findings underscore the importance of selecting an appropriate analytical method to study cloud properties between different cloud types.

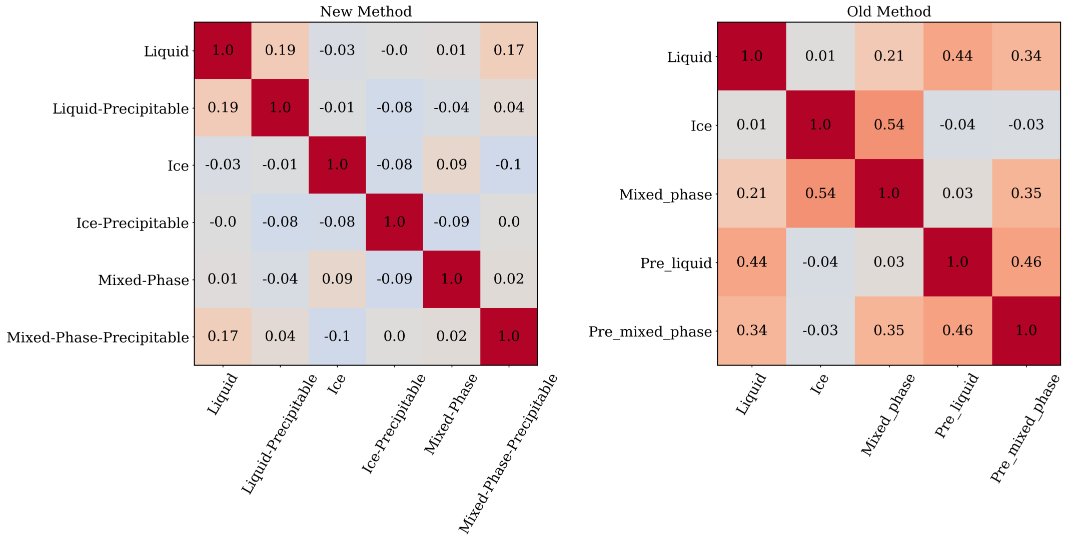
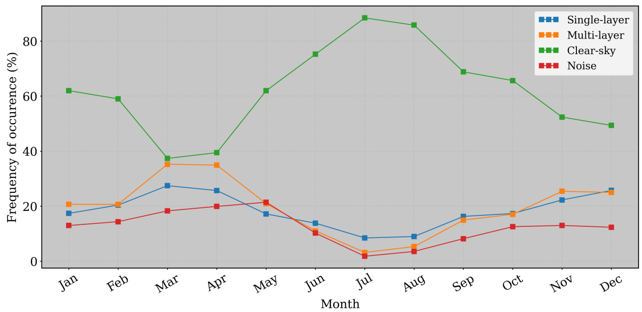


Figure 6. Correlation matrices comparing the daily occurrence between different cloud types using two methods. On the left, the New Method exhibits generally weaker correlations, indicating clearer distinctions between cloud types. In the right, The Old Method shows stronger correlations, suggesting more pronounced interdependence between cloud types

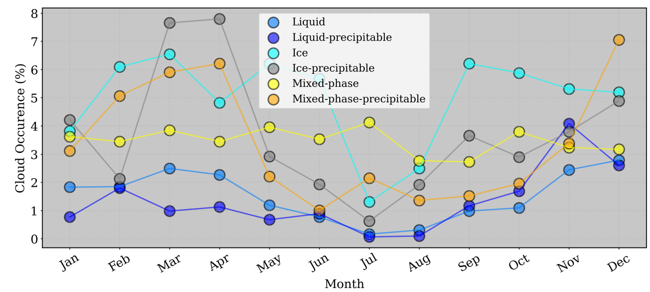
5 Results and discussion

5.1 Annual variability and daily cycle of clouds

Since the cloud classification by hydrometers grouping showed a more comprehensive classification, this algorithm is employed in the following analysis. This section aims to analyze the inter-annual and daily cycle variability of cloud frequency of occurrence at the Granada station. In this framework, a first overview on cloud occurrence is given by the left panel of Figure 7, which shows the monthly frequency of occurrence of different sky conditions, categorized into single-layer clouds, multi-layer clouds, clear sky, and noise (as explained in section 4.1). The data reveal a distinct seasonal pattern in cloud cover and clear sky occurrences. Clear sky conditions (green line) are predominant all year long, being most frequent in the summer months, peaking in July and August (> 80%), with a significant decline in spring, reaching the lowest point in March and April (40%). This pattern indicates a higher prevalence of clear skies during the warmer months and increased cloudiness during the colder months. Single-layer clouds (blue line) exhibit a relatively weak seasonal pattern throughout the year, with a slight peak in spring (35%) and a decrease in the summer months (10%), suggesting a moderate variation in single-layer cloud occurrences. Multi-layer clouds (orange line) show a similar pattern, with a peak in March and April and a reduction during the summer, indicating a seasonal preference for multi-layer clouds in the spring, when there is more multi-layer than single-layer. The noise category (red line) maintains a weak seasonal pattern throughout the year, comparable with multilayer in later spring and summer. Overall, the data highlight clear seasonal variations in sky conditions, with clear skies dominating in summer and increased cloudiness, both single and multi-layer, in the cooler months. Since this station exhibits a quite low cloud occurrence, a need to handle multilayer clouds is revealed to fill gaps of cloud properties at regions such as the western Mediterranean. Particularly, these multi-layer are problematic due to inaccurate LWP partitioning, leading to a misclassification of liquid pixels inside clouds, as observed by Tatiana Nomokonova et al.. Therefore, the following analyses are focused only on single-layer clouds, seeing their physical properties can be retrieved more accurately.



(a)



(b)

Figure 7. On the left, inter-annual frequency of occurrence for single layer clouds (blue), multi-layer clouds (orange), clear sky (green) and noise (red). Right panel, inter-annual single layer cloud occurrence for different cloud categories (see section 4.1). Each cloud types is indicated in the legend

Monthly frequency of occurrence is depicted in the right panel of Figure 7. Mixed-phase clouds (yellow) exhibit relatively consistent occurrences with slight fluctuations around 3% and 4%. Liquid-precipitable (dark blue) has a modest variation with zero occurrence in summer, followed by an increase during the transition to winter, reaching a maximum of 4% in November, suggesting increased precipitation during fall. Liquid clouds (blue) are more variable, showing a percentage of 3% in March and December, against zero in summer. Ice clouds (cyan) drop sharply in July and maintain almost constant levels around 5% and 7% throughout the rest of the year, indicating hotter temperatures that inhibit ice cloud formation during summer. Ice-precipitable clouds (gray) have a notable peak of 8% in spring, followed by a decrease in summer (1%) and an increase in fall, until they reach a local maximum of 5% during winter. It suggests seasonal variations in ice-related precipitation with maximum in spring, unlike the case of water-related precipitation, which showed a maximum in November. Mixed-phase-precipitable clouds (orange) peak in April (6%) and December (7%), indicating higher chances of mixed-phase precipitation towards spring and the end of the year. Overall, ice, precipitating ice, and precipitating mixed-phase clouds have the strongest seasonal pattern, indicating a higher sensibility of ice due to thermodynamic conditions than liquid droplets. Moreover, the precipitation in this region is intrinsically related to ice and mixed phase clouds.

During the summer temperatures are higher and just a few non-precipitating clouds were observed at our station. These clouds can be related to convection since we will observe that they have larger thickness and ice water content (IWC) at higher altitudes. Besides elevated temperatures, the Iberian Peninsula experiences a large low-pressure area over northern Africa, with high-pressure systems over the Atlantic Ocean and eastern and central Europe. This results in easterly winds over Andalusia, transporting moisture from the Mediterranean Sea in summer (F. J. Bello-Millán et al.). This combination of high temperatures and humidity can create an environment conducive to storm formation in Granada during the summer. In this context, the few clouds observed in summer are likely related to convectively formed cumulus or cumulonimbus clouds. This will be more evident when studying their physical properties in further sections.

Next, the daily cycle of cloud occurrence should uncover the influence of temperature around mid-day, when the solar radiance is more intense. Although we already discuss the influence of synoptic and local phenomena in cloud formation, convection clouds should appear in the diurnal cycle. Figure 8 displays the diurnal cycle per season, showing no convection for any type of cloud during summer, fall and winter. Despite that fall shows an increase of ice clouds around mid-day, they are probably cirrus clouds, since their median base height and thickness are 8.5 km and 1.5 km, respectively (see Figure 9).

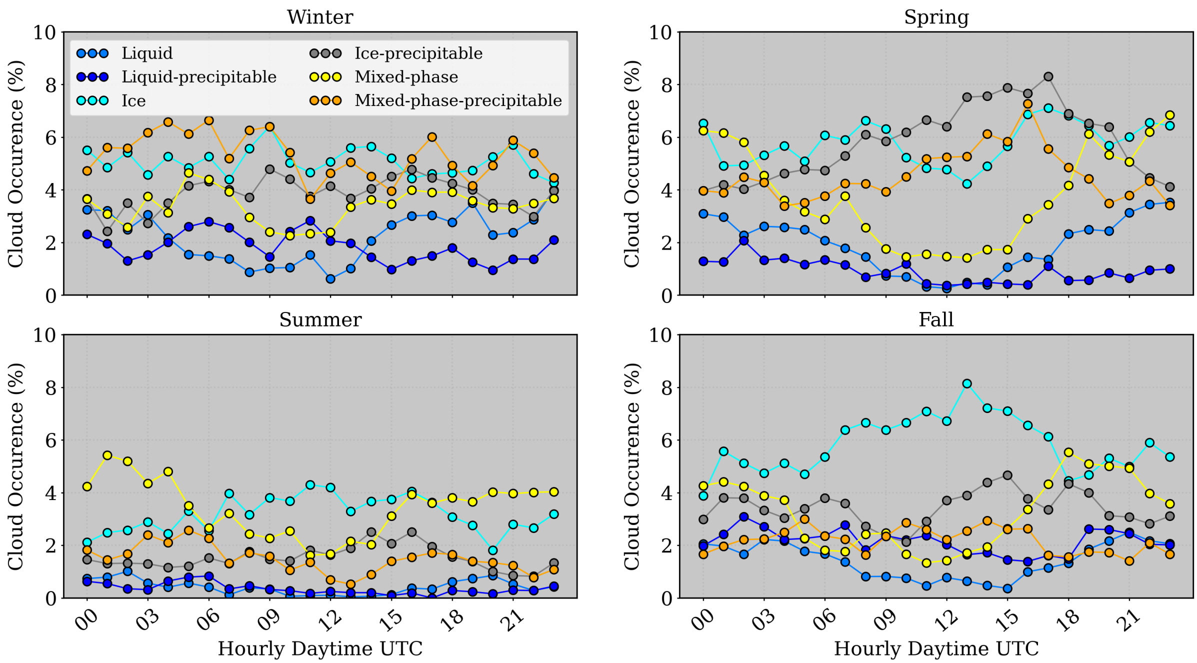


Figure 8. Daily cycle of cloud occurrence per season. The season is indicated in the title of each figure, where winter and spring are in the first row, and summer and fall in the second row. The color of each cloud category is indicated in the legend

However, spring shows a possible presence of convective clouds, with a slight peak of occurrence for Ice-precipitable and Mixed-phase-precipitable clouds between 09:00 and 21:00 UTC. Different from non-precipitating clouds which reached a minimum around mid-day, these clouds doubled the occurrence from 4% at 00:00 to 8% at 16:00-17:00 UTC. These observations are consistent with the findings of Shahi et al., where convection-permitting models of the Iberian Peninsula (IP) captured high values of mean precipitation during spring. Additionally, Alexandre Rios-Entenza et al. revealed that specific synoptic conditions, such as a reduction in the strength of the zonal circulation, can favor local and mesoscale convection during spring in the western region of the IP. This analysis confirms that the maximum cloud and rain occurrence observed during peak sunlight hours may be strongly associated with local convection.

5.2 Seasonal patterns in cloud base, top, and thickness

Seasonal evolution of cloud base height, top height and thickness are displayed in Figure 9. Each color represents a cloud type, where liquid clouds are in blue, ice clouds are in gray and mixed-phase clouds are in orange. Also, the lightest colors represent non precipitating clouds, and the darkest colors represent precipitation clouds. Thus, through cloud base analysis, precipitating clouds exhibit a significantly lower cloud base height (top panel) compared to non-precipitating clouds, as well as a narrower distribution. Regarding non-precipitating ice clouds, they possess the highest cloud base heights, with median values around 8 km. The ice clouds with base height above the 50/75th percentile can be normally associated with cirrus clouds. Next, non-precipitating mixed-phase clouds have the second highest cloud base, showing considerable variability, except during summer, when values stabilize at 5.5 km. Also, these clouds present a higher base in fall, where the 50th percentile reaches 6.5 km. Differently, the precipitating mixed phase clouds have much less variability and higher cloud base at 4.5 km in summer, as expected. For non precipitating liquid clouds, cloud bases are highly variable during summer, achieving significantly different heights compared with other seasons. Overall, in summer, precipitating mixed-phase clouds, and ice precipitating clouds demonstrate elevated cloud base heights, attributed to the relatively high temperatures and low humidity at this season. Liquid clouds also have elevated base height in summer, however only a few cases were registered, with 50th and 75th percentiles around 4 km and 6 km, respectively. These clouds can be related to stratocumulus and stratus clouds, which were dis-coupled from cumulonimbus at the final stage in summer.

Following, precipitating clouds also have significantly lower median cloud top height (panels at the center of Figure 9) compared to non-precipitating clouds and exhibit a narrower cloud top distribution, except in the case of ice clouds. Non-precipitating ice clouds have median cloud tops around 9-10 km, while ice-precipitable clouds demonstrate highly variable cloud tops. Given their well-defined cloud bases, these clouds exhibit greater cloud thickness, indicative of storm activity at our station. The cloud tops for liquid clouds mirror the variability of their cloud bases, especially during summer. These clouds exhibit a well-defined cloud thickness, with non precipitating liquid clouds having a thickness around 300 m and precipitating liquid clouds reaching up to 500 m. Non precipitating mixed-phase cloud tops also present high variability. Nevertheless, they displayed a well-defined cloud thickness, with similar distribution along seasons. Their median cloud thickness is around 400 m, increasing to 500 m for the precipitating case (dark orange), with slightly higher values in summer. In general, non-precipitating mixed-phase clouds have median cloud tops around 7 km during spring, summer, and winter, but reach 8.5 km in fall. Then, precipitating mixed-phase clouds have cloud tops around 3 km in spring, fall, and winter, peaking at 7 km in summer.

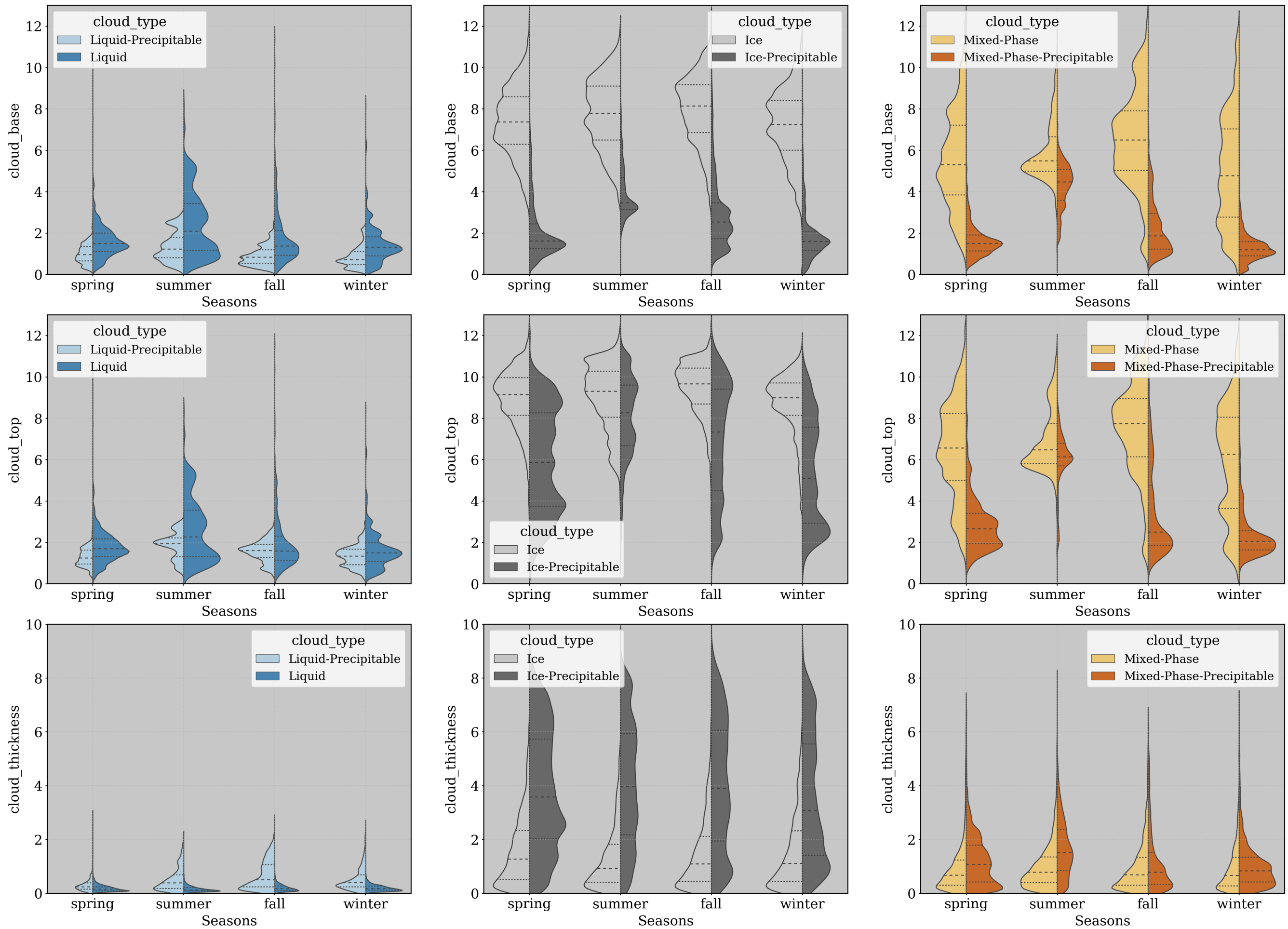


Figure 9. Seasonal evaluation of cloud base height, top height and thickness. The cloud bases are in the first row, cloud tops in the second row and cloud thickness in the third row. Also, each cloud type is indicated in the legend, where liquid clouds are in the first column, ice clouds in the second column and mixed-phase clouds in the third column. Finally, the dashed lines are indicating the 25/50/75th percentiles

Overall, ice-precipitable clouds exhibit the greatest cloud thickness, followed by ice, mixed-phase-precipitable, mixed-phase, liquid-precipitable, and liquid clouds. It is important to note that, despite Cloudnet post-processed data being corrected for liquid attenuation, cloud tops for precipitating clouds with higher thickness can be underestimated due to high liquid attenuation.

5.3 Cloud microphysics properties

5.3.1 Liquid clouds

The monthly median profiles of LWC shows high variability of LWC in function of height for liquid clouds (Figure 10, right panel, marked by hot colors and constant lines of 1000 mg m^{-3}). For particular months such as February, March and April, a notably variability below 1 km and within 2.5 and 4 km (right panel of Figure 10) is observed. Nevertheless, above 500 m, really high values observed in the monthly profiles are not representative of the entire season as can be seen in the left panel of Figure 10. This figure shows no clear pattern with altitude at any season, LWC is generally presenting a random fluctuation around to 50 mg m^{-3}. Differently, below 500 m, a sharp peak of about 500 mg m^{-3} can be clearly identified in spring. This peak is related to liquid clouds during the springs of 2020, 2021 and 2022 (not shown), at this period. Except for the near surface largest LWC observed in spring, no clear differences were observed between seasonal profiles. Following, summer exhibited a lack of liquid phase clouds, with some elevated LWC at higher levels as shown in the left panel of Figure 10.

Next, the average liquid water path (LWP) (bottom panel of Figure 10, black star) shows a considerable variability of LWP from January to June (also early fall). It highlights a very wide variety of liquid clouds in these months. Also, a notable increase in the median LWP occurs in April, aligning with their maximum cloud occurrence. From August to October, there is a gradual increase in average LWP (marked by the black star), while the median stays stable. This trend corresponds with the increased occurrence of liquid clouds from summer to fall, impacted by different mechanisms of cloud formation. Since these months reported just a few occurrences of liquid clouds, as can be checked by the blue line, which shows the number of cloud profiles per month, their statistics will be much more sensible to these mechanisms. As a result LWP in summer is the smallest, and there is no significant difference between the average and the median.

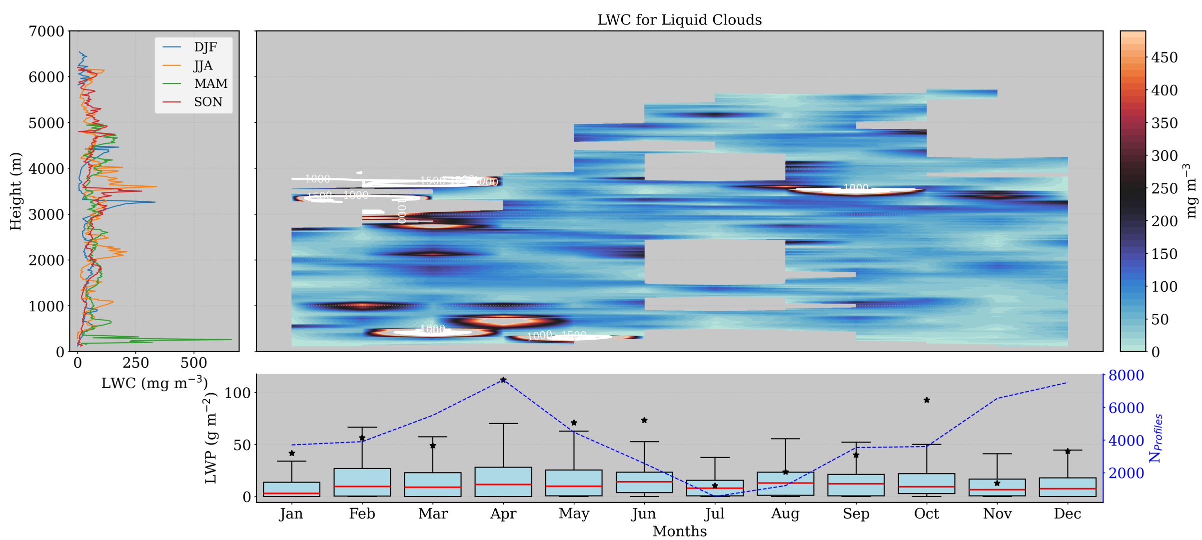


Figure 10. Seasonal and monthly average of liquid water content (LWC) and liquid water path (LWP) for liquid clouds. Left panel shows the seasonal median profiles of LWC. Right panel shows the monthly median profiles of LWC and monthly box-plot of LWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

In terms of droplet effective radius, liquid clouds are well consistent throughout the year, showing small differences in terms of median values (top panel of Figure 11), mostly ranging between 5 and 9 μm. There is no discernible pattern in r_{liq} across seasons, with higher altitudes weakly correlating to larger droplet sizes. In addition, there are almost no liquid droplets above 4 km, just a few occurrences in summer and early fall were registered.

However, the r_{liq} is generally stable around 5 μm, with a small decrease in July, when almost no clouds were registered. The seasonal profiles confirm the uniformity in r_{liq} distribution with height, showing minimal variation with altitude. It suggests a consistent cloud composition and droplet size throughout different atmospheric conditions. These findings underscore the stable nature of liquid clouds in terms of droplet size and LWC. Normally, these clouds have a small thickness, do not presenting a wide range of r_{liq}

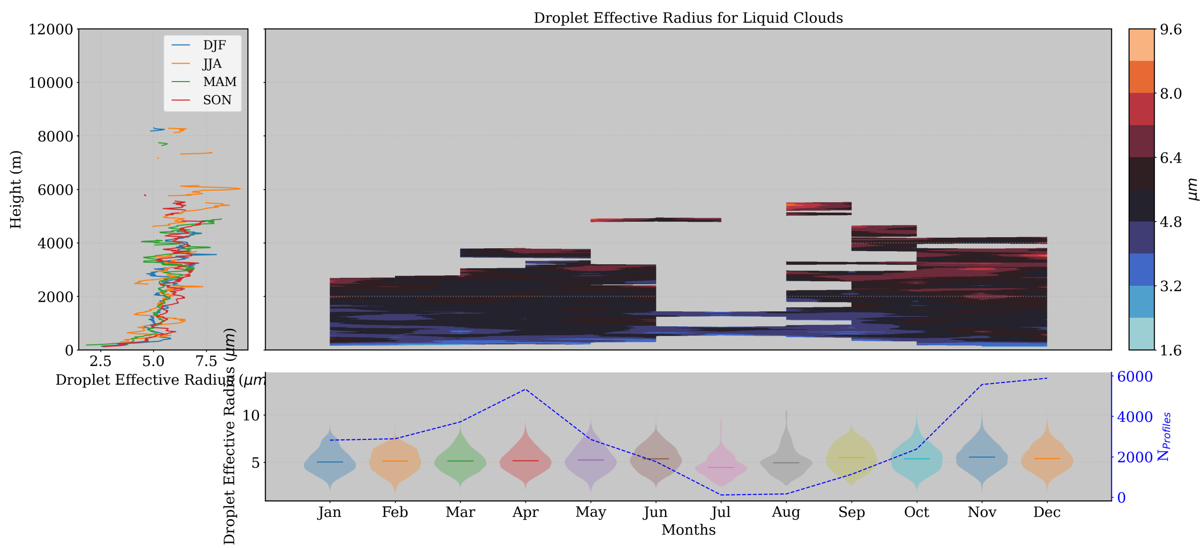


Figure 11. Seasonal and monthly statistics on droplet effective radius (r_{liq}) distribution for liquid clouds. Left panel shows the seasonal median profiles of droplet effective radius. Top-right panel shows the monthly profiles of the median r_{liq}, while the bottom-panel displays the size distribution of r_{liq} by months.

For precipitating liquid clouds, significant liquid water content (LWC) is primarily found below 3 km (top-right panel of Figure 12), consistent with their median top heights. Next, regions with significant values of liquid water content show broader areas, which is expected as these clouds are thicker than purely liquid clouds. Monthly median profiles indicate that higher amounts of liquid water content occurs in later fall, winter and spring. However, the highest amount of LWC is identified in June (early summer), very close to the ground (300 m), which is related to only a few cases of these clouds registered in summer (see the blue line in the bottom panel). Then, in the rest of summer and early fall, atmospheric water content is almost zero. In spring, significant values are found even closer to the surface, explained by relatively humidity months in 2020, 2021, and 2022 as already observed for non precipitating liquid clouds. At the last, winter shows a peak of LWC of about 250 mg m^{-2} at 2 km.

Overall, the seasonal profiles of LWC emphasizes a clear increase of LWC in function of height during fall, winter and summer. For the last, two peaks of 250 mg m^{-3} are clearly identified at 2 and 3 km, respectively. This season presented more variability with height, showing more peaks of LWC in function of height. Next, fall and winter also presented a peak of about 250 mg m^{-3} at 2 and 2.5 km, respectively. In terms of LWP, Figure 12 below indicates substantial variability of LWP in January, suggesting a diverse variety of liquid clouds during this month. In contrast, spring exhibits relatively consistent LWP across its months, reflecting a stable period for precipitating liquid clouds. In these months, their median stays below 50 g m^{-2}, and summer has even less LWP, except for June, as previously observed.

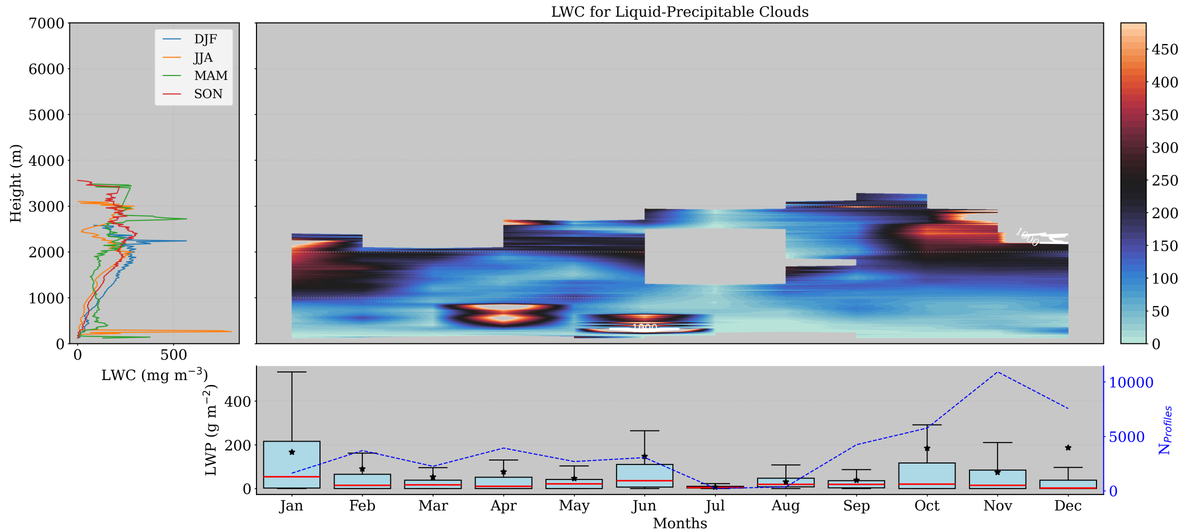


Figure 12. Same as Figure 10, but for precipitating liquid clouds.

Different for the purely liquid clouds, precipitating liquid clouds demonstrated radically different seasonal profiles of median r_{liq} (see Figure 13). In spring and summer, the r_{liq} is smaller than fall and winter the closer to the ground, as can be seen by light blue and warm colors. In addition, r_{liq} in fall are the largest, often around 15 μm below 2 km. These sizes are normally related to drizzle droplets, which are more present in this season. The highest median r_{liq} on the ground in the left panel of Figure 13 is also related to drizzle particles. This panel also shows that summer and winter exhibit similar r_{liq} values up to 2 km of altitude, around 10 μm, decreasing to approximately 5 μm at 4 km. On the other hand, spring shows an increase of r_{liq} from 5 μm to a peak of 15 μm around 3 km altitude.

Their monthly size distribution in the bottom panel displays different size distributions over the year, where higher median values are found in fall as already observed, specially in November. During spring and early summer, the median is maintained around 10 μm, then decreases in summer, and increases again in fall as expected.

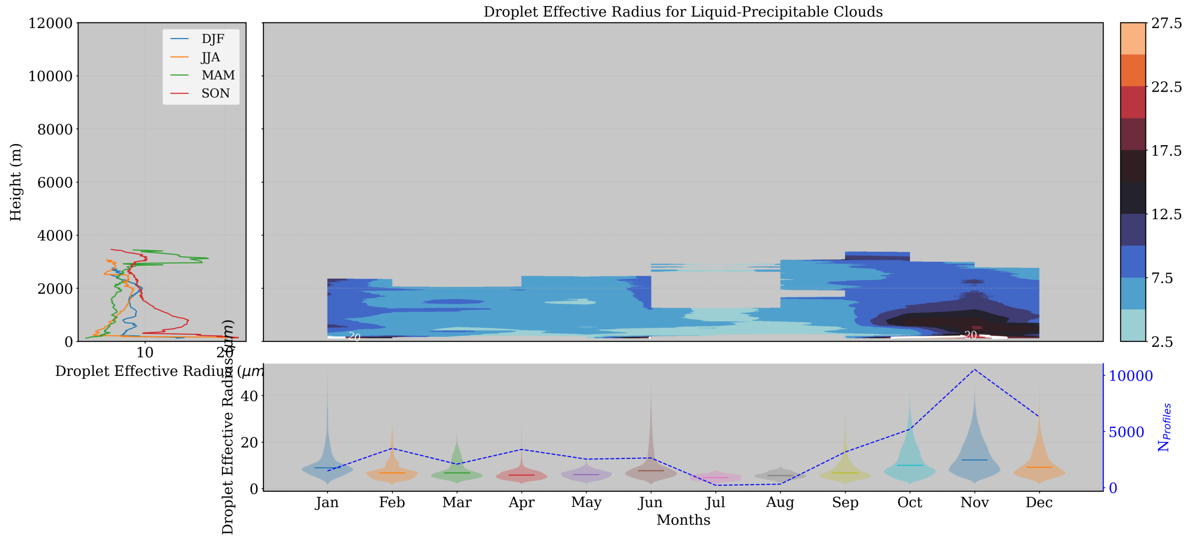


Figure 13. Same as figure 11, but for precipitating liquid clouds.

Liquid clouds are challenging to analyze due to their low frequency of occurrence. They exhibit significant variability with height, making it difficult to discern seasonal differences in liquid water content (LWC) profiles, except at low levels. Near the surface, LWC is much higher in spring, though their droplet size (r_{liq}) remains relatively constant around 5 μm. In contrast, precipitating liquid clouds display a more pronounced seasonal pattern, with higher LWC and larger r_{liq}. Notably, the presence of drizzle particles near the surface is associated with larger r_{liq}. During summer, these clouds exhibit two distinct peaks in LWC in function of height, one at 2 km and another at 3 km

5.3.2 Mixed Phase clouds

Next, mixed-phase clouds indicate generally lower liquid water content (LWC) values compared to liquid-phase clouds (top-right panel of Figure 14). These clouds have a consistent occurrence throughout the year (around 3-4%), but exhibit very high variability of LWC in function of height, as expected since their cloud base also has such variability. Especially in summer and winter, high LWC are encountered above 8 km, surpassing 1000 mg m^{-3}. As for the liquid cloud case, these values occur just a few times at this level, and do not affect the seasonal median (left panel of Figure 14), since other summer-months have much smaller LWC at this height.

Although LWC varies greatly during winter, spring and fall, with most pronounced occurrences below 4 km, their integral remains really close to zero over the year. This can be observed by the bottom panel of Figure 14, which shows an absence of LWP throughout the entire year in terms of the median. On the other hand, just a few cases with elevated values are generating high mean LWP, highlighting the presence of outliers. Next, vertical profiles of median LWC by season denoted high variability below 4 km and above 10 km, for all stations. Between this level, LWC is relatively small at all seasons, fluctuating around zero. Otherwise, median LWC have local maximums around 100 mg m^{-3}, with two notable peaks surpassing 500 mg m^{-3}

at 500 m during spring and summer (same case as for liquid clouds). These seasonal variations underscore the complex and dynamic nature of mixed-phase cloud formation and their differing behaviors across various atmospheric conditions.

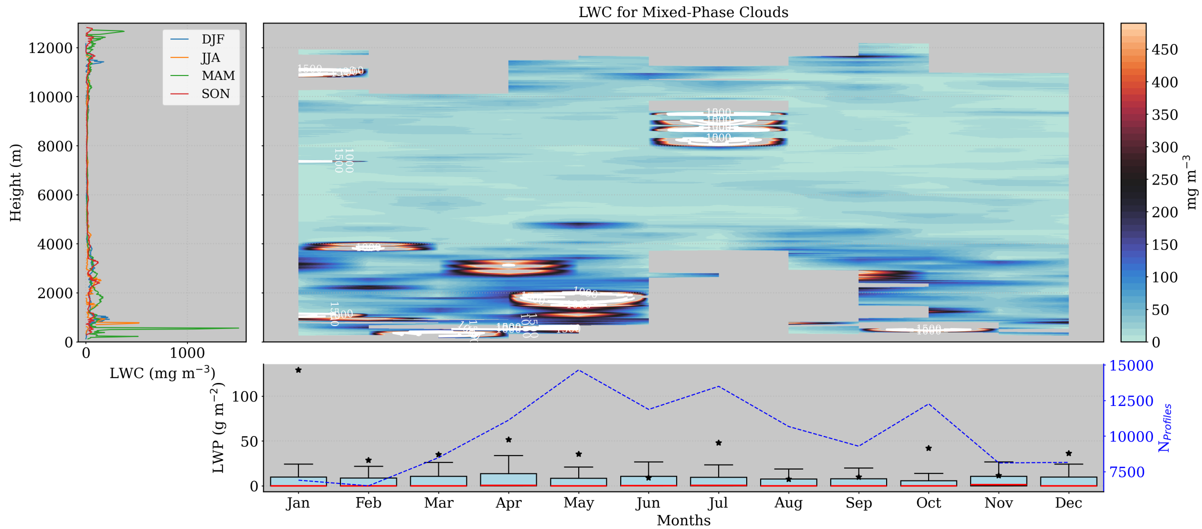


Figure 14. Same as Figure 10, but for mixed-phase clouds.

In terms of droplet effective radius (r_{liq}), a totally different nature is observed in relation to LWC. The last has expressive amounts of liquid water at low and high layers (< 4 km and > 10 km), while the first is exhibiting high droplet sizes at middle levels. Large values of droplet sizes can be observed on the right-top panel of Figure 15 in dark blues and hot tones. They are normally located between 4 and 8 km throughout the entire year, except for July, which also shows larger droplets at 10 km. As already discussed, liquid water is not common at those levels, but when detected, are followed by high mass concentration, suggesting the detection of cumulus clouds. These clouds can have supercooled liquid water at high levels.

Overall, summer presents higher droplets compared to colder months, with an effective radius between 14-22 μm at medium heights (> 4 km and < 6 km), and between 12-18μm around 10 km. On the other hand, droplet sizes are generally close to 10 μm inside these clouds. This is shown by the monthly distribution of r_{liq}, where the modes are located at even smaller values than the median. In fact, cloud droplets are slightly larger in summer and have a good probability to reach 15 μm, but in general they present smaller droplets.

In addition, there is a clear dependency of droplet size with the height. Seasonal profiles on the left panel of Figure 15 show droplet size increasing up to 5 km, then starting to decrease. Thus, it is followed by all seasons, indicating that particular seasonal differences of droplet size is only grasped by monthly profile analysis. However, the seasonal profile highlights a general behavior, which can be explained by the ice particle formation. Since temperature decreases with height, ice particles will be formed, increasing the water vapor competition, stopping the process of growing of liquid droplets to form ice particles as temperature continues to decrease. These factors will decrease the size of water droplets and increase the mass concentration of ice particles. The last can be confirmed by the left panel of Figure 16.

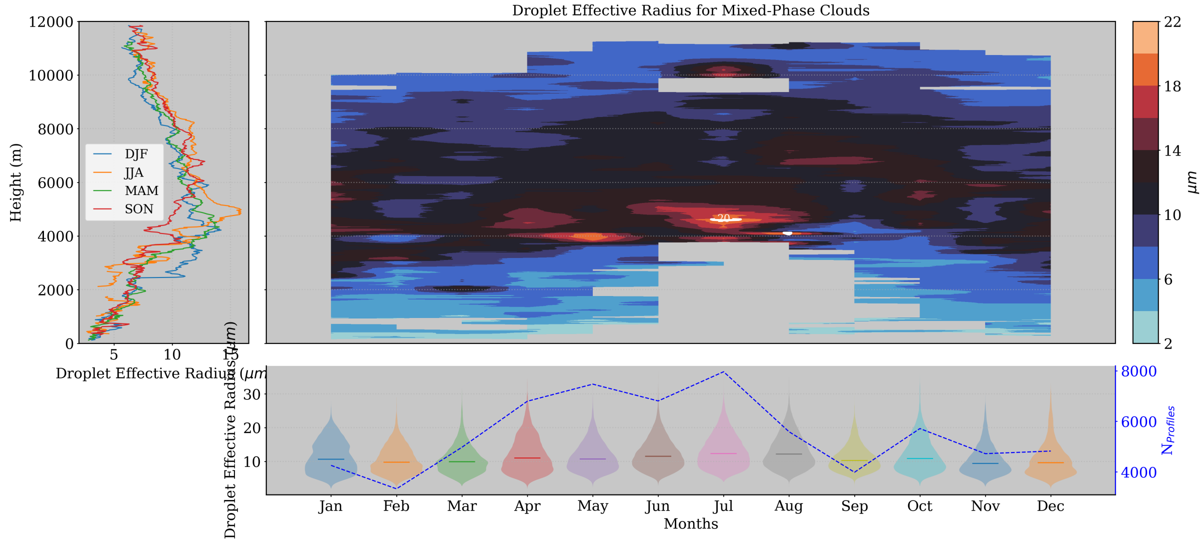


Figure 15. Same as figure 11, but for mixed-phase clouds.

Regarding the ice water content, it exhibits significant variation with height, especially during summer. The monthly median profiles in Figure 17 shows that ice mass is greater at middle/high altitudes, as expected. Normally, significant values of IWC of about 20-40 mg m^{-3} are found between 4 and 10 km, except for July. As for the LWC, July also presented elevated IWC above 10 km (more than 60 mg m^{-3}), complementing our hypotheses that these clouds can be classified as cumulus clouds. Cumulus clouds are the first stage of cumulonimbus clouds and can have considerable amounts of ice, especially at their top. In addition, these clouds have a considerable thickness as those observed for mixed phase clouds (some of them can reach 2 km of thickness).

In terms of IWP, these clouds have low ice amounts as can be seen by the median in the bottom panel, normally below 10 g m^{-2} throughout the entire year. However, the presence of just a few cases with high amounts of ice (outliers) can pull the average up. It happens specially in April and summer, where the physical properties of these clouds are more variable. Also, as already observed, IWC is increasing with altitude, generally formed slightly above 2 km, increasing until 7 km for colder months (winter and spring) and 8 km for warmer months (fall and summer). Winter and fall follow practically the same function of height, as well summer follows the same behavior of fall up to almost 8 km. Above this level, summer has the largest amount of IWC, with maximum, varying between 20 and 10 mg m^{-3}.

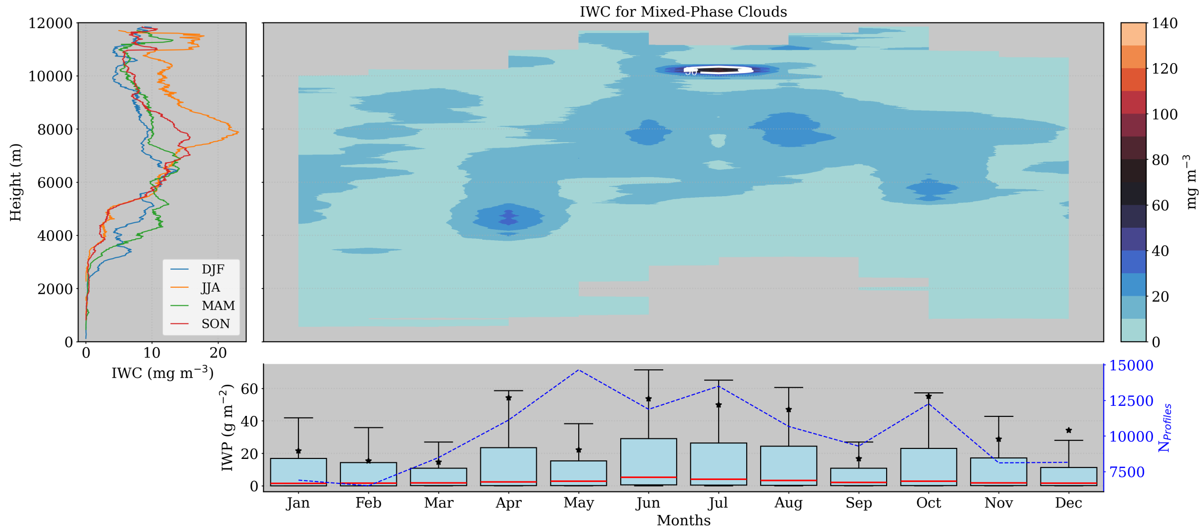


Figure 16. Seasonal and monthly average of ice water content (IWC) and ice water path (IWP) for mixed-phase clouds. Left panel shows the seasonal median profiles of IWC. Right panel shows the monthly median profiles of IWC and monthly box-plot of IWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

In terms of effective radius, their size follows almost the inverse of the IWC in function of the height. Figure 17 undoubtedly shows that higher radii are at low levels, except in July where it's possible to find r_{ice} larger than 50 μm above 10 km. Hence, the general properties of these clouds can be described as having larger ice particles with low ice water content (IWC), particularly at lower altitudes. This characteristic is not limited to summer, during which these clouds are distinctly different from those observed throughout the rest of the year. In addition, the monthly distribution (bottom panel of Figure 17) shows that larger ice effective radii are more frequent than small particles. The median is normally fluctuating around 45 μm, with the mode above this value, indicating high probability of more elevated radius. The seasonal profiles (on the left panel of Figure 17) also strengthened our finding that effective radius decreases with height. Actually, it is still almost constant with height up to 6 km, where it starts to decrease. Its values in summer are slightly higher but there are no significant differences over the seasons. Normally, the r_{ice} is around 30 μm above 10 km, except for July.

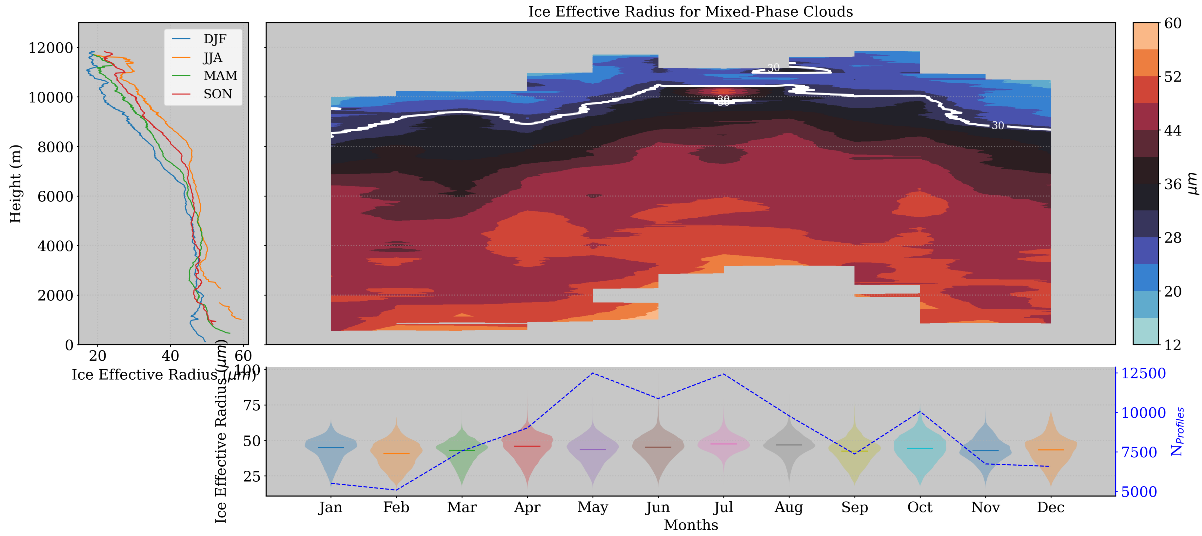


Figure 17. Seasonal and monthly statistics on ice effective radius (r_{ice}) distribution for mixed-phase clouds. Left panel shows the seasonal median profiles of ice effective radius. Top-right panel shows the monthly profiles of the median r_{ice}, while the bottom-panel displays the size distribution of r_{ice} by months.

Regarding the LWC, precipitating mixed-phase clouds follow a similar pattern as non precipitating clouds. However, for the precipitating case, significant amounts of liquid water are located at lower altitudes (< 4km), and they present a stronger seasonal pattern (Top-right panel of Figure 18). It is natural, considering that we already observed a strong seasonal variability in their frequency of occurrence. In addition, their liquid water content is normally higher than precipitating clouds, as expected. It is worth mentioning that the color scale in Figure 18 is the same as the non precipitating case in Figure 14, which shows much lighter areas (low values of LWC). Moreover, It is evident that these precipitating clouds do not present such high variability with height, in agreement with their cloud base distribution that showed much less variability than the non precipitating case. Following, winter and spring showed the highest values of LWC, surpassing 1000 mg m^{-3} at some cases close to the surface. These periods show really sharp peaks of LWC, also observed during fall, below 1 km (see left panel). It shows that high concentrations of water in liquid phase are normally found at low levels, following the temperature pattern as expected. It is very known that the temperature at our region significantly increases in summer and fall, allowing liquid cloud droplets to be formed at higher levels. However, these months also have less humidity in the atmosphere, hindering the cloud formation as already observed by the less cloud occurrence.

For summer, a water shortage is clearly observed, especially at low levels. Both in summer and fall (hot months), considerable amounts of LWC are present at medium levels in the atmosphere (> 2 and < 4.5 km). This can be easily seen in the left panel of Figure 14, where fall (in red) and summer (orange) have almost the same vertical profile, varying around 200 mg m^{-3} between 2 and 4.5 km. At the same layer, the median profile of winter and spring goes through zero at 3 km. Furthermore, the integrated liquid water path has high variability in colder months, with medians below 50 mg m^{-2} and averages above 100 mg m^{-3}. In summer, there are just a few occurrences of these clouds, resulting in less variability, with both medians and averages close to zero.

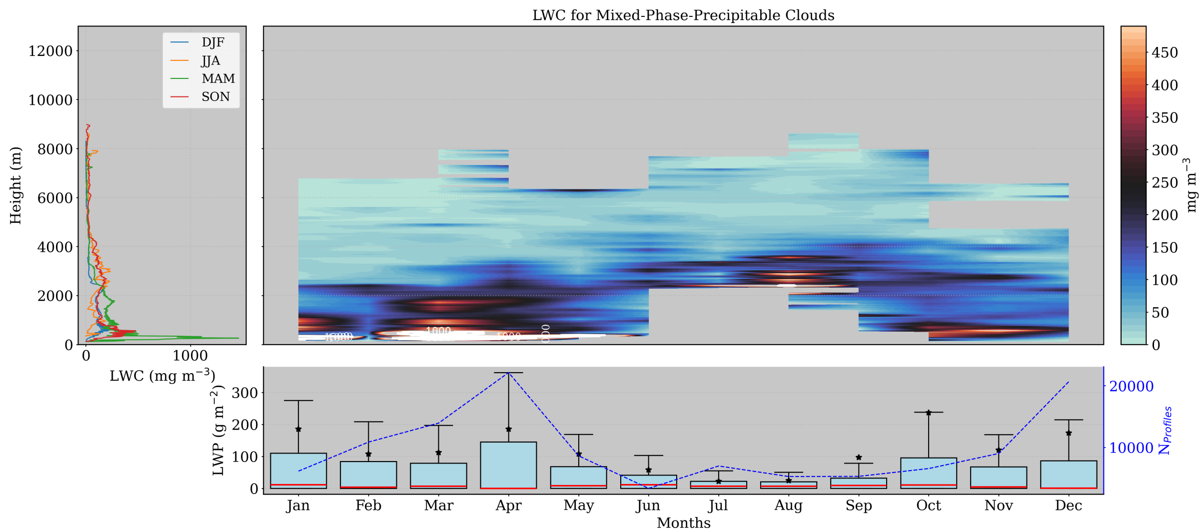


Figure 18. Same as Figure 14, but for precipitating mixed-phase clouds.

In relation to the ice effective radius (r_{ice}), values below 8 km range from 48 to 64 μm, with slightly larger particles observed at specific heights during winter and summer—around 5 km in winter and 7 km in summer. Notably, in summer, larger ice particles of 70 μm can be identified at 10 km, coinciding with higher ice water content (IWC). Such high values can be attributed to strong convective motions, supporting the hypothesis that these summer clouds are cumulonimbus. At the same altitude, r_{ice} values in other months are generally smaller than 30 μm, except in fall, which also presents higher values at 12 km. Above 1 km, particle size begins to increase (since large particles below are associated with drizzle below the cloud base). For winter and spring, r_{ice} increases up to 5 km (reaching 50 μm) and then decreases to 20 and 30 μm, respectively. In contrast, for summer and fall, r_{ice} increases up to 7 km (reaching 50 μm), then decreases before increasing again to reach second maxima of 50 and 63 μm, respectively. However, despite the larger variability with height, the monthly distribution of r_{ice} shows a highly variable pattern with a sharp mode at the median position of 50 μm.

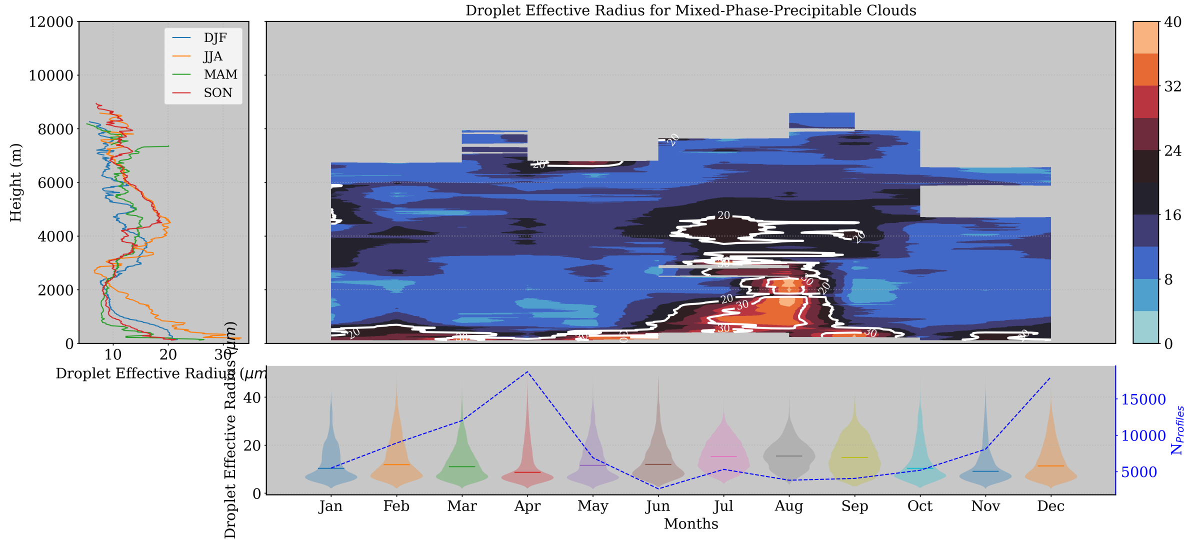


Figure 19. Same as figure 15, but for precipitating mixed-phase clouds.

Following, Figure 20 reveals that the ice amount inside these clouds is consistently above 3 km, exhibiting high variability with height. The largest amounts of ice water content (IWC) are found during January and November around 5 km, and in June at approximately 8-10 km. However, the median vertical profile by season (left panel of Figure 20) shows no peaks at these levels for winter, indicating that the high IWC observed in January represents particular cases rather than typical winter conditions. Typically, winter and summer IWC ranges around 20 mg m^{-3} at 3.5 km and fluctuates around 5 mg m^{-3} from 4 km to 9 km. The low frequency of occurrence of these clouds in summer and fall results in comparable amounts of ice across the months, with fewer outliers. This leads to local maxima in the seasonal profiles of IWC (left panel) that align with the monthly maxima (right panel). Summer shows an almost constant IWC around 20 mg m^{-3} from 5.5 to 7.5 km, with a sharp peak of 200 mg m^{-3} at 9 km. Fall also exhibits IWC around 20 mg m^{-3}, but consistently from 5 to 9 km. Particularly, July, August, and September have the most comparable largest amounts of IWC in the atmosphere. Consequently, the 25th, 50th, and 75th percentiles during these months are relatively higher, with 25% of the data showing IWC values exceeding 200 mg m^{-2}.

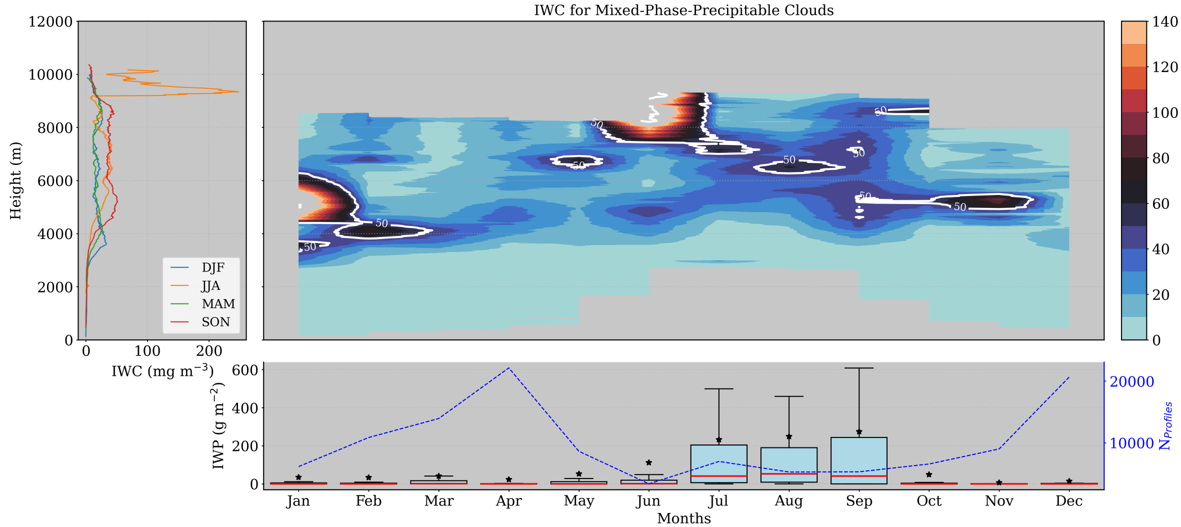


Figure 20. Same as Figure 16, but for precipitating mixed-phase clouds.

In terms of effective radius, the largest values are found below 8 km, with the highest values occurring in summer and fall (dark red), as shown in Figure 21. During these months, ice particle sizes exhibit less variability with height, ranging around 50-55 μm. Conversely, winter and spring show variability between 45 and 55 μm. The left panel of Figure 21 clearly separates warm months (summer-fall) from colder months (winter-spring). In warm months, the ice effective radius remains almost constant around 50 μm up to 8 km. Above this altitude, particle sizes decrease in fall, while summer exhibits a sharp maximum at 9 km, coinciding with the maximum IWC region. Winter and spring also have similar r_{ice} values to those in summer and fall up to 4 km, after which the sizes decrease with height, reaching 20 μm at 10 km. Despite this variability of r_{ice} with height, the monthly distribution shows that within these clouds, r_{ice} is generally around 55 μm.

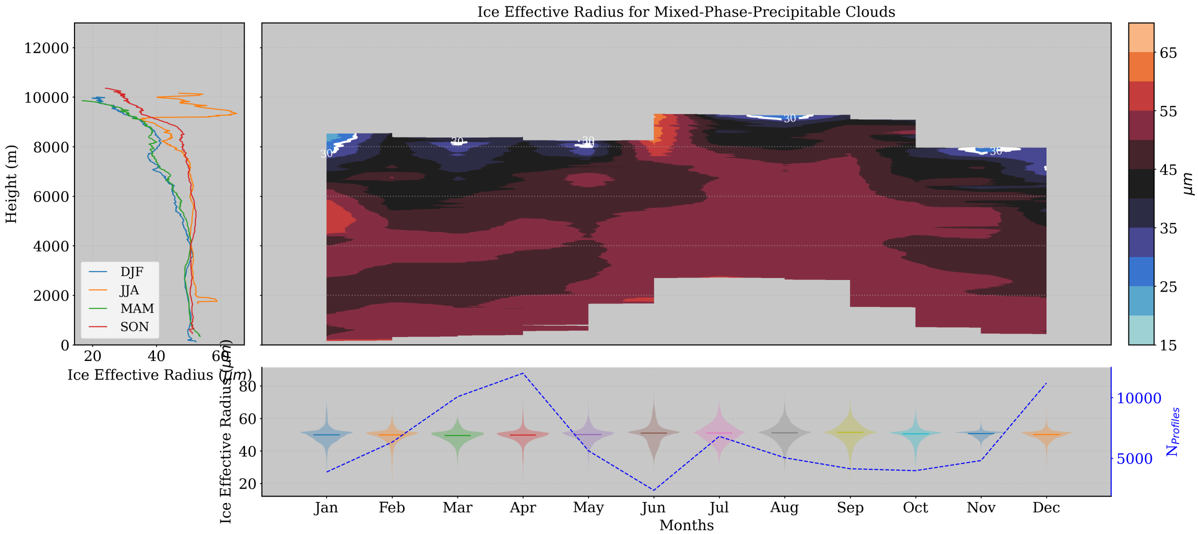


Figure 21. Same as figure 17, but for precipitating mixed-phase clouds.

Overall, Mixed-phase clouds exhibit less liquid water content (LWC) than liquid clouds. Typically, the LWC in liquid clouds is concentrated at low levels, with some exceptions in June and July, where high LWC values are observed above 8 km. Mixed-phase clouds also contain less ice water content (IWC) compared to ice clouds, with the IWC found at higher altitudes due to dependency on low temperatures, leading to elevated IWC levels higher in the atmosphere during summer. The droplet effective radius in mixed-phase clouds is larger at mid-levels, while the ice effective radius is larger at mid-levels during winter and spring and around 8 km during fall and summer. Unlike this pattern, the ice effective radius in these clouds slightly decreases with height up to 6 km, where the rate of decrease intensifies. Precipitating mixed-phase clouds exhibit a stronger seasonal profile, with higher LWC, particularly in spring and fall at lower levels. They also show significantly more IWC in summer and early fall. The droplet effective radius (r_{liq}) in precipitating mixed-phase clouds is typically higher than in mixed-phase clouds, especially near the surface and at mid-levels, due to the presence of drizzle particles at low levels. In terms of the ice effective radius (r_{ice}), these clouds display a clear seasonal variation, with higher r_{ice} values in summer and fall above 4 km. Additionally, precipitating clouds show larger IWC and r_{ice} values at 9 km in summer, suggesting a possible association with cumulus clouds.

5.3.3 Ice Clouds

The ice water content (IWC) of ice clouds exhibits clear seasonal patterns, as evident from Figure 22. The smallest ice amounts are observed in summer, particularly in July, coinciding with the minimum frequency of ice cloud occurrence. In contrast, the highest IWC values, exceeding 50 mg m^{-3}, are found at altitudes of 4-6 km in June and October, and near the ground in January. Typically, these clouds are situated higher in the atmosphere, with cloud bases around 7-8 km, with only 25% of ice clouds having bases below 6 km. These lower clouds generally contain the largest ice amounts, ranging between 30 and 100 mg m^{-3}, except in January. Notably, January exhibits much higher IWC near the ground, an anomaly attributed to extreme weather events such as the Filomena storm in 2020 and the Gloria storm in 2021, which significantly impacted the western Mediterranean region (Samuel T. Buisán et al.; María Eugenia Pérez-González et al.; Marta de Alfonso et al.). Conversely, clouds above 8 km, often associated with cirrus clouds, typically have much lower ice content, ranging from 0 to 20 mg m^{-3}, except in fall, where clouds with IWC around 50 mg m^{-3} are observed at 12 km. These seasonal variations are also evident in the left panel of Figure 22, which shows substantial ice near the surface in winter. This was already identified as the influence of two snowfall events, resulting in a median IWC of 400 mg m^{-3} around 1 km. The variability compared to other years without such high IWC values is highlighted in the bottom panel of Figure 22, where the average liquid water path (LWP) is much higher than the median for January, indicating that typical IWP values for January are low, around 10 mg m^{-2}. Overall, the monthly IWP distributions for ice clouds are relatively consistent throughout the year, indicating a well-defined annual IWP pattern, except in January and July. January shows lower occurrence rates and is impacted by the snowfall events, while July is characterized by an extremely low occurrence of ice clouds, resulting in a minimum LWP.

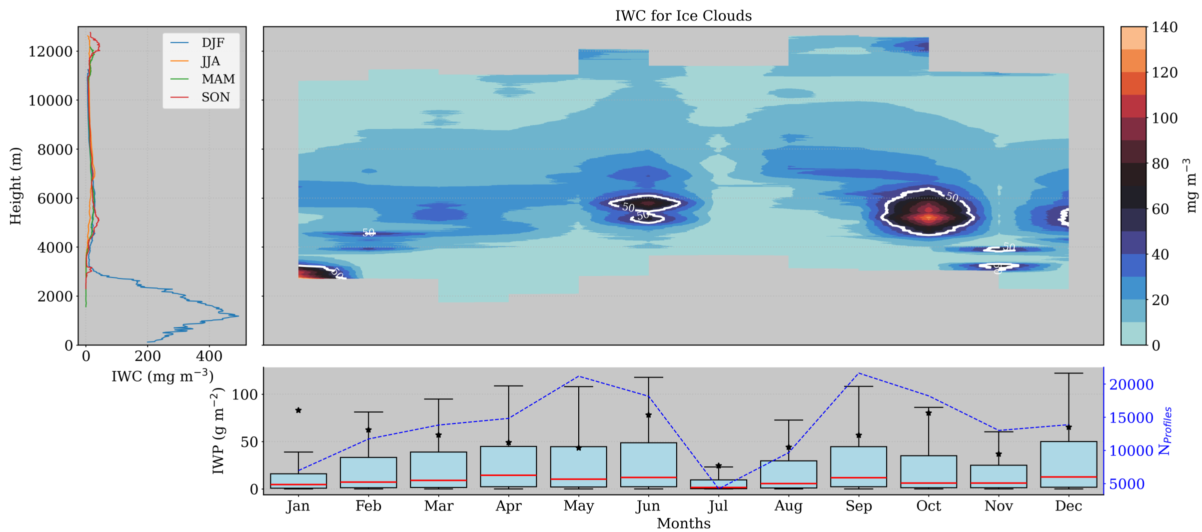


Figure 22. Seasonal and monthly average of ice water content (IWC) and ice water path (IWP) for ice clouds. Left panel shows the seasonal median profiles of IWC. Right panel shows the monthly median profiles of IWC and monthly box-plot of IWP in the bottom. For the boxplots, medians are denoted by red lines and the average by black stars.

In terms of ice effective radius, their size is also larger at middle levels, ranging around 50 and 60 μm. In addition, January also presented elevated r_{ice} closer to the ground (clearly seen in the left panel of figure Figure 23), as a result of the snowfalls. However, beyond this month, the maximum size of 60 μm is normally observed exclusively in June and October in mid levels. Left panel of Figure 23 shows that above 6 km, ice particle size diminishes, indicating a larger cloud radius at lower levels, a pattern also observed in mixed-phase clouds. Typically, ice particles are larger at low and mid-levels, except in clouds with significant convective activity, such as cumulus and cumulonimbus clouds. Cirrus clouds situated above 8 km have an ice effective radius between 40 and 45 μm, decreasing with altitude up to 11 km. At this level, during spring and summer, these clouds reach 25 μm, while in winter, the size reduces to 20 μm (see Left panel of Figure 23). In fall, a slight increase in size is observed at 12 km, coinciding with high IWC levels, linked to cirrus clouds with higher ice crystal concentrations. The monthly distribution reveals a stable median ice particle size of approximately 38 μm throughout the year, indicating that this value is generally representative when considering all radii within the atmosphere for this cloud type. However, January presents a distribution with a new mode around 50 μm, as expected due to the snowfall events already discussed.

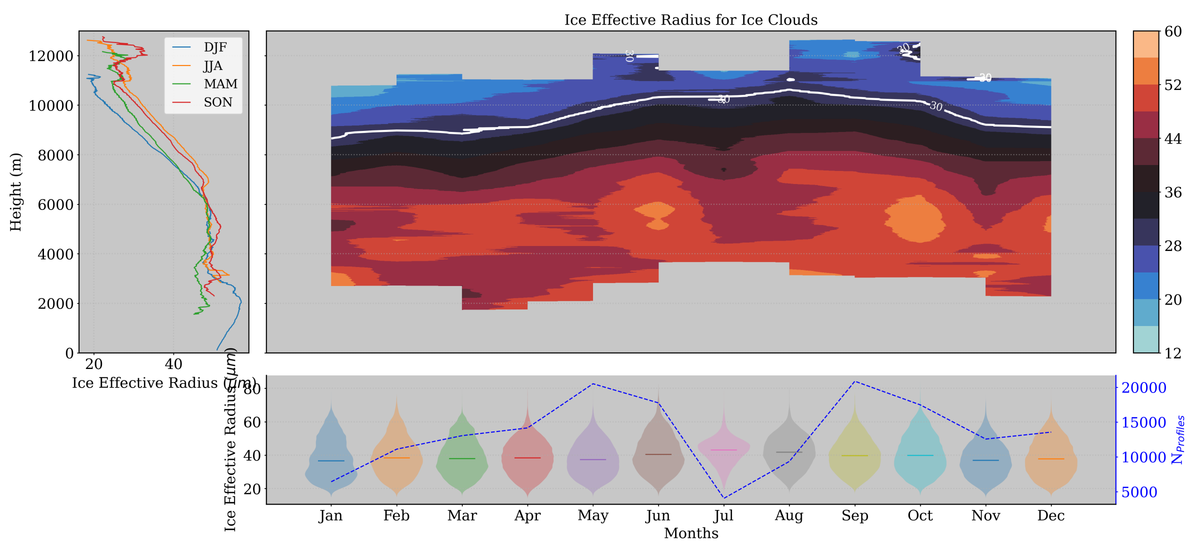


Figure 23. Seasonal and monthly statistics on ice effective radius (r_{ice}) distribution for ice clouds. Left panel shows the seasonal median profiles of ice effective radius. Top-right panel shows the monthly profiles of the median r_{ice}, while the bottom-panel displays the size distribution of r_{ice} by months.

Ice precipitating clouds are the most frequent at our station, characterized by their greater thickness and lower cloud base compared to non-precipitating ice clouds. Similar to precipitating mixed-phase clouds, these clouds possess a well-defined melting layer that separates the cloud base from rain droplets and drizzle. The highest concentrations of ice, exceeding 50 mg m^{-3}, are found between 2 and 5 km, with slightly elevated levels during summer and fall. Notably, regions with ice concentrations over 200 mg m^{-3} were identified around 6 km in June and September, and between 9 and 12 km in July. This significant ice presence at high altitudes was not observed in ice clouds, though a relatively high ice content was detected in precipitating mixed-phase clouds in June, suggesting a correlation with the high ice amounts in summer and thicker clouds. The substantial ice amounts at high altitudes in summer are strongly linked to convective motions, providing evidence that precipitating ice clouds in summer may be associated with cumulonimbus clouds. Additionally, these clouds may represent a subsequent stage of precipitating mixed-phase clouds, exhibiting a similar pattern but with increased thickness and ice water content, particularly in summer. The left panel of Figure 24 illustrates similar median profiles of IWC in summer and fall, with summer displaying a secondary maximum around 11 km and fall showing a more pronounced second maximum at 13 km. In contrast, spring and winter exhibit similar maxima of 100 and 150 mg m^{-3} around 4.5 km, respectively, slightly below the primary maxima of 200 mg m^{-3} observed in summer and fall around 6 km. The bottom panel of Figure 24 highlights that these clouds contain significantly higher ice water content compared to other clouds, especially during summer and fall, with the 75th percentile increasing from 500 to 1000 mg m^{-2}. Furthermore, it is noteworthy that the IWP in other months is much lower, indicating that in months with a higher occurrence of these clouds, the IWP is more variable and generally smaller. The lower occurrence of warm clouds results in comparable IWC values and reduced variability.

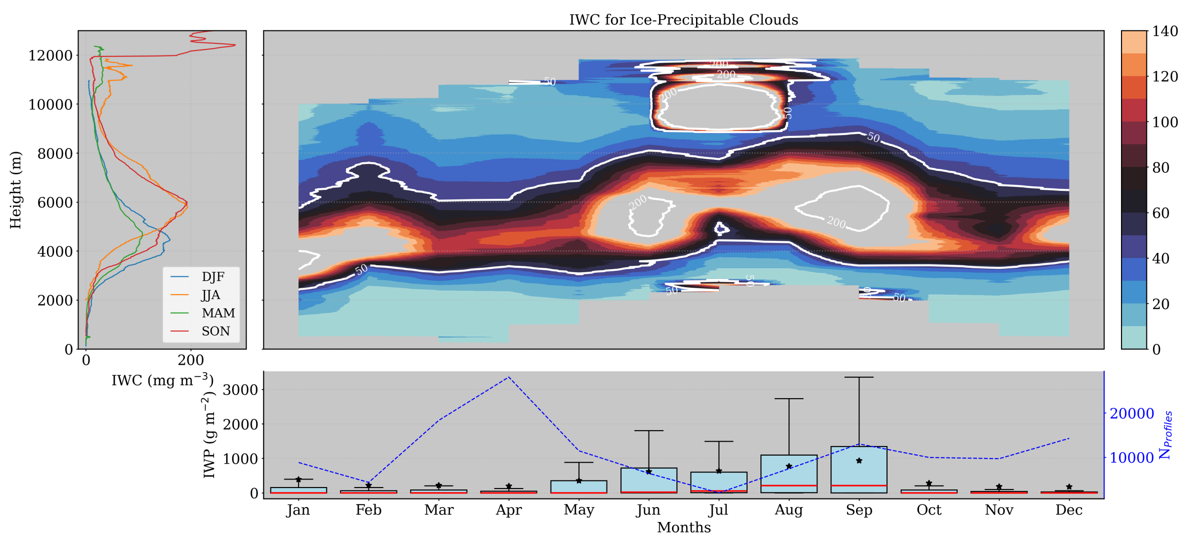


Figure 24. Same as Figure 22, but for ice clouds.

In relation to the ice effective radius (r_{ice}), values below 8 km range from 48 to 64 μm, with slightly larger particles observed at specific heights during winter and summer—around 5 km in winter and 7 km in summer. Notably, in summer, larger ice particles of 70 μm can be identified at 10 km, coinciding with higher ice water content (IWC). Such high values can be attributed to strong convective motions, supporting the hypothesis that these summer clouds are cumulonimbus. At the same altitude, r_{ice} values in other months are generally smaller than 30 μm, except in fall, which also presents higher values at 12 km. The left panel of Figure 25 depicts a distinct behavior of the median profiles compared to other clouds. Above 1 km, particle size begins to slowly increase (since large particles below are associated with drizzle below the cloud base). For winter and spring, r_{ice} increases up to 5 km (reaching 50 μm) and then decreases to 20 and 30 μm, respectively. In contrast, for summer and fall, r_{ice} increases up to 7 km (reaching 50 μm), then decreases before increasing again to reach second maxima of 50 and 63 μm, respectively. Despite the larger variability with height, the monthly distribution of r_{ice} shows a highly variable pattern with a sharp mode at the median position of 50 μm.

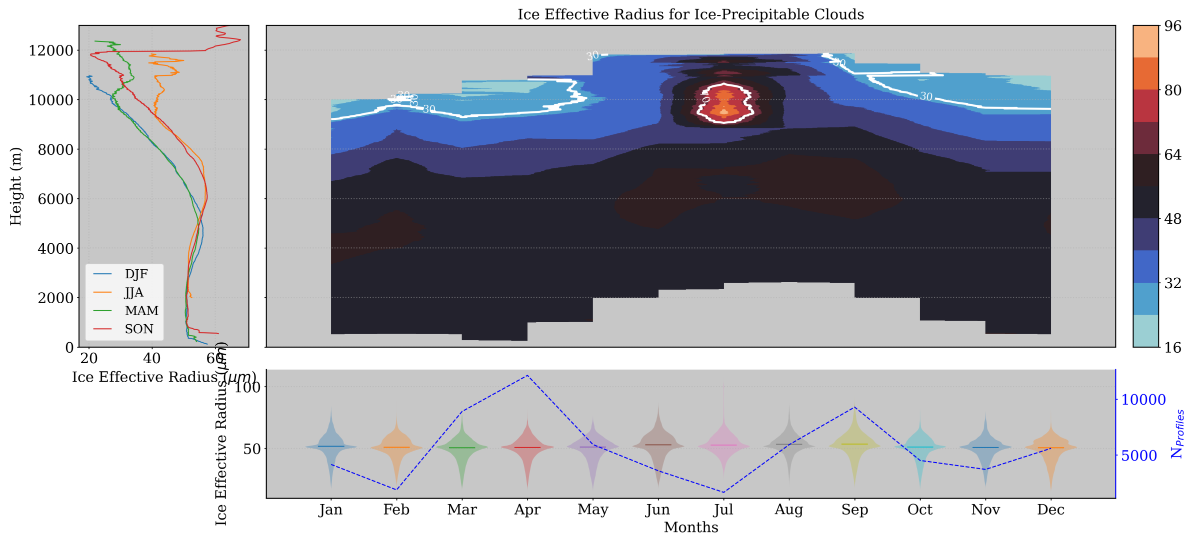


Figure 25. Same as Figure 23, but for mixed-phase clouds.

In general, these clouds exhibit larger amounts of ice water content (IWC) at mid-levels (4-8 km), particularly in early summer and fall, while other summer months show an ice shortage. In January, most high IWC values were recorded during extreme snowfall events, such as the Filomena and Gloria storms, resulting in high median IWC at low levels during this month. The r_{ice} is also influenced by these snowfalls, showing higher values close to the surface. Typically, r_{ice} ranges around 50 μm up to 6 km, where it starts to decrease, indicating smaller ice particles in higher-level clouds. Precipitating mixed-phase clouds display a much stronger seasonal pattern in their physical properties. Summer and fall show a peak of IWC at 6 km, while winter and spring exhibit a less pronounced peak at 4 km. These clouds contain significantly more IWC than non-precipitating ice clouds, and they are markedly different, with greater thickness. Similar to precipitating mixed-phase clouds, they have large amounts of IWC and r_{ice} at high levels in summer (> 8 km). Additionally, they demonstrate clear seasonal differences in r_{ice}, which is greater in summer and fall

6 Conclusions

This study presents a statistical analysis of cloud properties using remote sensing ground-based instruments at the AGORA ACTRIS CCRES station, marking the first such analysis in Spain employing the Cloudnet database. Utilizing over half a decade (2018 to 2023) of post-processed data from the Cloudnet consortium, we classified single-layer clouds and retrieved their physical properties using the target classification product. However, this product exhibited some misclassification at our station, necessitating adjustments to avoid misinterpretation of the results. To enhance cloud classification, we compared previous time-based cloud classification methods from recent studies with a novel approach that categorizes clouds into ice (precipitating ice), liquid (precipitating liquid), and mixed-phase (precipitating mixed-phase) clouds. Evaluating this classification method is crucial for understanding the derived cloud properties, as these will be influenced by the chosen method. Consequently, using this new method, we conducted an inter-annual analysis alongside a diurnal cycle analysis of single-layer clouds to assess seasonal variability and identify convective clouds. Additionally, monthly and seasonal analyses of macrophysical and microphysical properties were presented by cloud type to elucidate their differences.

Appendix A: Cloud assessment by time

This method have been used by recent studies such as Răzvan Pîrloagă et al.; Tatiana Nomokonova et al.; Stefan Kneifel et al.. Each study has adopted some different criteria to classify single layer clouds, but they all classify cloud layers by profile. Here, we adopted some criteria based on these studies and employed the following method: cloud layers in the atmospheric column were classified through pixel sequence inquiry. These pixel sequences were divided in the following 5 groups according to Răzvan Pîrloagă et al.: liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only an "ice" pixel sequence was encountered. Mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified whether at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Unlike previous studies, whether rain pixels are found with more than 20 pixels (256-1022 m) of distance from cloud base, the clouds are not classified as precipitable, assuming that rain droplets are not falling from these cloud layers. In all cases, for being considered a cloud layer, a minimum sequence of 3 cloud pixels (38-153 m) (Tatiana Nomokonova et al.) must be satisfied, in addition to a minimum sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" to separate multi-layers. Therefore, free hydrometeor layers in the neighborhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on Răzvan Pîrloagă et al., but these values are not currently well defined, their determination has been guided by empirical experience.

Appendix B: Pixel dilatation

The dilation method consists in expanding the ones in the cloud mask to connect the closest groups of hydrometeors, which are likely to belong to the same cloud. The dilatation follows the scheme below, where hydrometeor pixels are represented in blue. Left image represents the original hydrometeor pixels where two groups of clouds were identified. Thus, a dilatation of 1 pixel is applied in all directions, represented by the red pixels in the right image. Therefore, the dilation process will unite these groups with at least 2 pixels of separation distance. Since they are connected, the cloud classification algorithm will save the coordinates of these cloud pixels in just one group (not two, as previously identified).

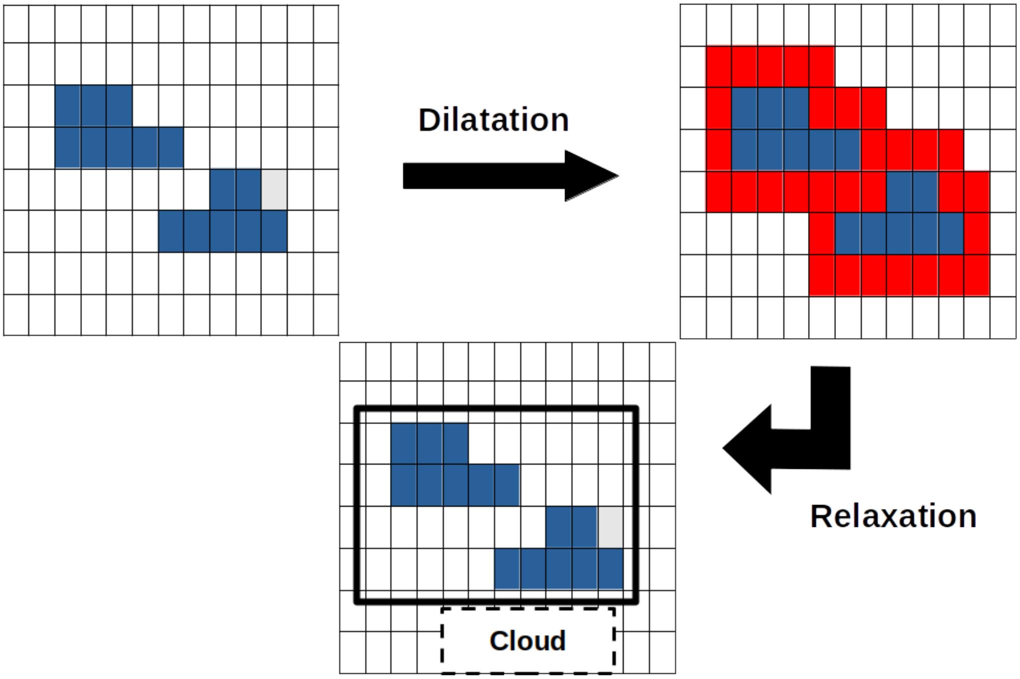


Figure B1. Pixel dilatation and pixel contraction illustration. Left panel shows two hydrometeor groups. Right panel still shows these two clouds and the dilated pixels in red. Since these clouds are connected, the classification algorithm will understand there is only one cloud. Bottom panel shows the result of the pixel contraction, returning cloud pixels to its original position

Next, a contraction process is done to return the coordinates of true hydrometeors to the original position. The contraction will remove the coordinates of fake hydrometeor pixels (pixels in red on the above figure). Afterwards, the classification algorithm can follow its chain and calculate the percentage of each hydrometeor type inside each group to classify different clouds in the cloudnet product.

Acknowledgements. TEXT

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